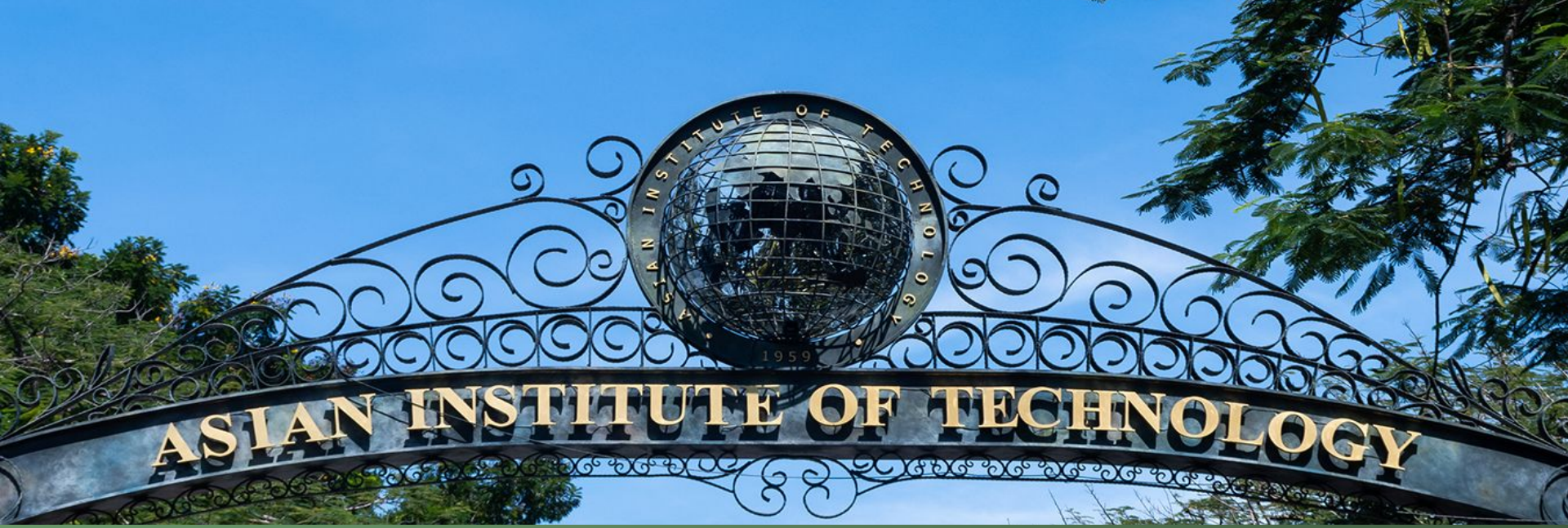



# Hybrid Convolutional Neural Network-Vision Transformer Architecture with Parallel Feature Processing and Cross-Modal Attention for Lung Segmentation

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PMDS 03
December -2025

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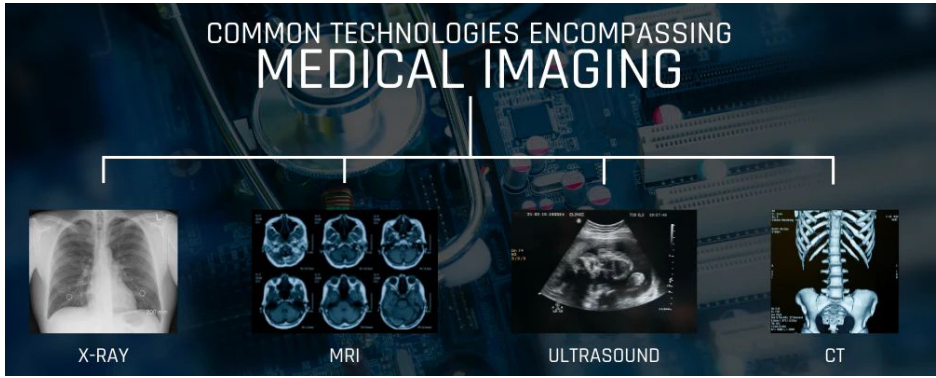
# Agenda

- Introduction
- Literature Review
- Methodology
- Experiment Result
- Conclusion

# Introduction

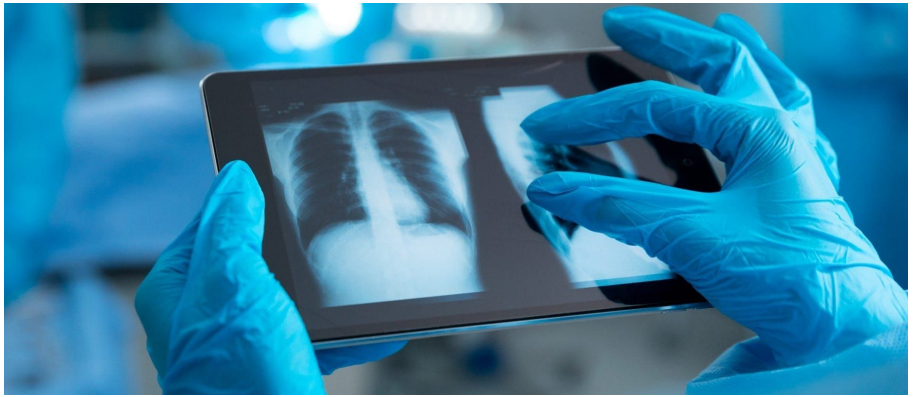
# Medical Image Analysis in Modern Health Care

- **Medical image analysis**: Automated detection & diagnosis across healthcare systems
- **Chest radiography dominance**: Most common exam (~33% of all radiological procedures)
- **Clinical importance**: Cost-effective, low radiation, wide pathology detection capability
- **Computer vision challenge**: Bridging technical innovation with clinical requirements



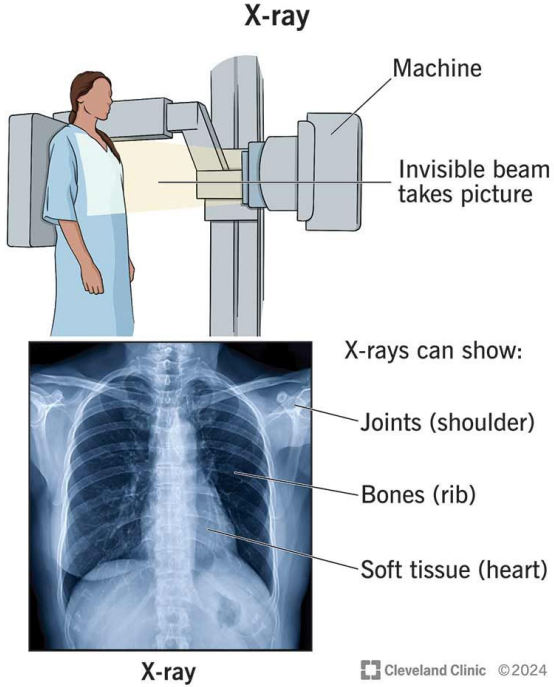
# Benefits of Lung Segmentation

- **Clinical foundation:** Essential prerequisite for automated chest X-ray diagnosis systems
- **Quantitative capabilities:** Enables lung volume measurement & disease progression monitoring
- **Disease detection gateway:** Critical for pneumonia, tuberculosis, lung cancer identification
- **Diagnostic reliability:** Segmentation accuracy directly determines Computer-aided diagnosis (CAD) system performance
- **Healthcare impact:** Supports radiologist workflow & enables screening in resource-limited settings



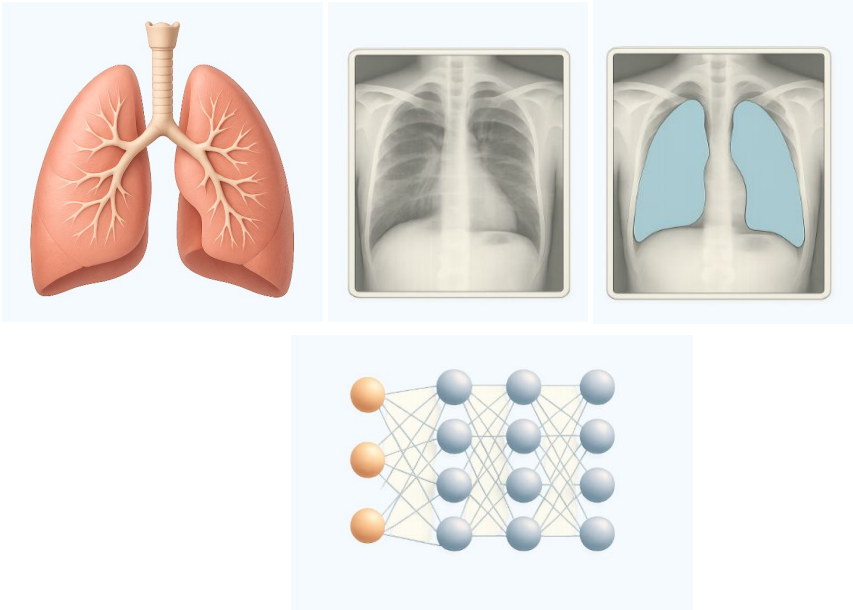
# Challenges Of Chest X-ray Image Capturing

- **Projection complexity:** 3D anatomical structures collapsed onto 2D imaging planes
- **Structural interference:** Rib cage shadows intersect with lung field boundaries
- **Anatomical occlusion:** Cardiac silhouette & diaphragm obscure lung portions
- **Pathological variability:** Pleural effusions, consolidations alter expected appearances
- **Population diversity:** Age, gender, health status create significant anatomical variation
- **Technical factors:** Image quality, positioning, equipment differences add complexity



# From Traditional Methods to Deep Learning Revolution

- **Traditional era (pre-2015):** Rule-based methods, manual feature engineering, domain expertise
- **CNN breakthrough :** U-Net (2015) architecture enabled automatic feature learning & local precision
- **Vision Transformer emergence (2020):** Self-attention mechanisms for global context modeling
- **Hybrid potential:** Classification tasks show CNN-ViT combinations achieve superior performance
- **Critical gap:** Limited exploration of hybrid approaches for segmentation-specific applications
- **Segmentation challenge:** Requires both fine boundary detection AND global context coherence



# Research Gap & Motivation

- **Current hybrid focus:** on disease classification & recognition tasks
- **Segmentation requirements:** Simultaneous need for local precision & global anatomical understanding
- **Architecture limitation:** Pure CNN approaches miss global context; pure ViTs struggle with fine details
- **Research opportunity:** No effective CNN-ViT fusion specifically optimized for boundary detection
- **Clinical motivation:** Improved segmentation accuracy could enhance diagnostic reliability

# Literature Review

# From Manual to Hybrid Models

| Timeline | Method | Strength | Gap |
|---|---|---|---|
| Pre-2015 | Thresholding, edge detection, morphological operations, K-means, SVM, KNN | Fast, interpretable | Manual tuning, poor generalization |
| 2015-2020 | CNN/U-Net | Automatic features, good boundaries | Limited global context |
| 2020-2021 | Vision Transformers | Global understanding | Limited boundary details |
| 2021-Now | Hybrids | Best of both approach | Lung Segmentation focus |

# CNN Revolution

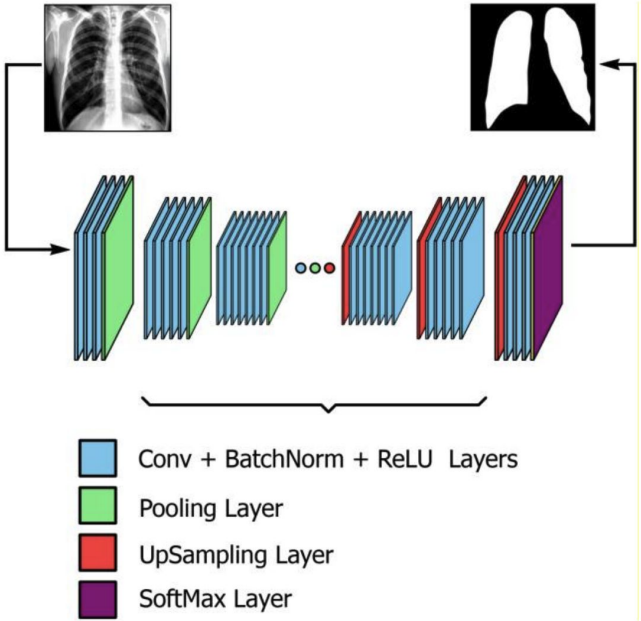
Converting classification models (VGG, ResNet) to segmentation, automatic feature learning vs manual engineering

**Key Strengths:**

- **Hierarchical feature learning:** Low-level edges → High-level anatomical structures
- **Translation invariance:** Robust to patient positioning variations
- **Local spatial patterns:** Excellent for rib detection, boundary identification
- **Transfer learning:** ImageNet pre-training accelerates medical adaptation

**Remaining Limitations:**

- "Limited modeling capabilities for long-range dependencies"
- Need very deep networks for global context
- Gradient Vanishing in deep architectures
- Missing Global lung shape understanding




Architecture of Deep Encoder-Decoder Convolutional Neural Networks - 2016

# U-NET (2015)- The Architecture Defined Medical Segmentation

**Innovation:** "De facto choice" for medical segmentation

- Symmetric encoder-decoder with skip connections
- Combines "deep feature extraction with pixel-level segmentation"

**Key Strengths:**

- Skip connections preserve fine details
- Multi-scale processing
- ISBI 2015 Cell Segmentation Challenge winner

**Remaining Limitations:**

- Still CNN-based, lack of global context
- No adaptive attention mechanisms

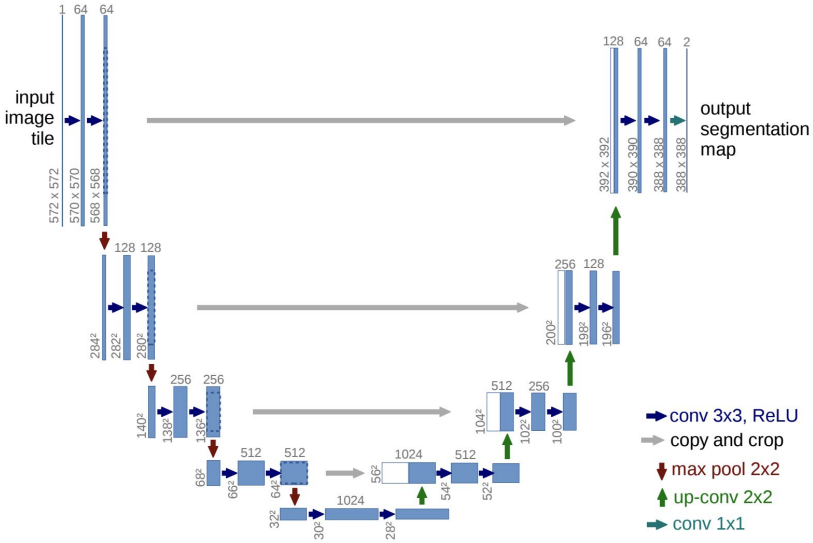



**Fig. 1.** U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

# Vision Transformer Emergence (2020)

**Transformer Advantages:**

- **Global receptive field:** Every patch attends to every other patch
- **Long-range dependencies:** Capture anatomical relationships
- **Self-attention mechanisms:** Learn importance of different regions
- **Semantic understanding:** Better grasp of overall lung structure

**Medical ViT Performance:**

- Strong results in global context understanding
- Better semantic comprehension of anatomical structures

**Segmentation Limitations:**

- **Struggle with fine-grained boundary details**
- **Patch-based processing:** May miss pixel-level precision
- **High computational cost**

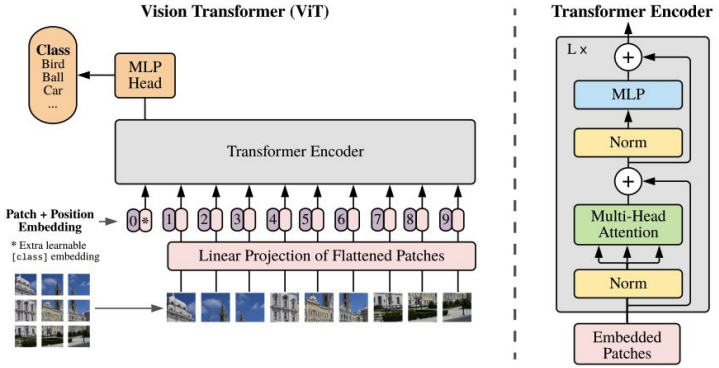



Figure 1: Model overview. We split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder. In order to perform classification, we use the standard approach of adding an extra learnable "classification token" to the sequence. The illustration of the Transformer encoder was inspired by Vaswani et al. (2017).

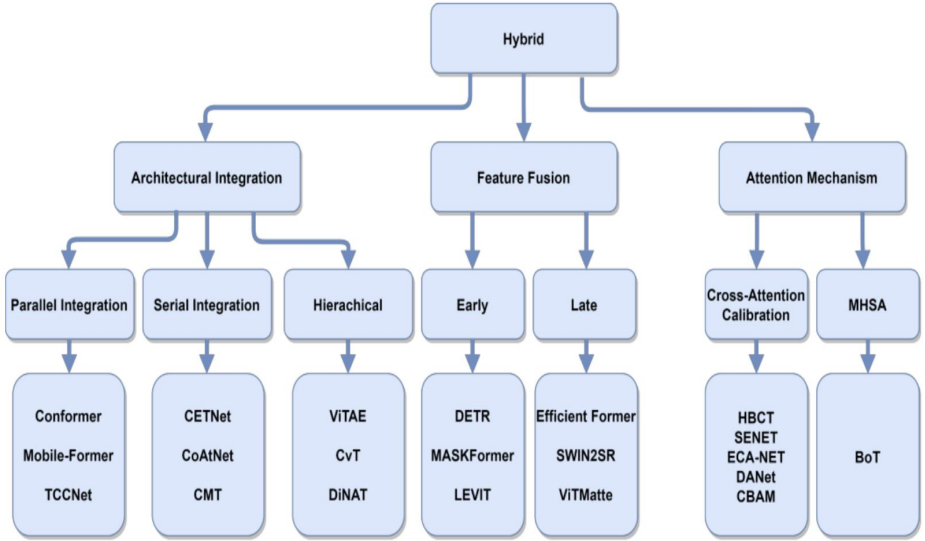
# Hybrid CNN-Transformer Approaches

**"Best of Both Worlds: CNN + Transformer"**

**Question:** How to effectively combine these architectures => Fusion Strategies

**Two Main Approaches Emerged:**

1. **Sequential:** TransUNet (CNN → Transformer)
2. **Parallel:** TransFuse (CNN || Transformer → Fusion)




Yunusa, H., Qin, S., Chukkol, A. H. A., Yusuf, A. A., Bello, I., and Lawan, A. (2024). Exploring the synergies of hybrid cnns and vits architectures for computer vision: A survey. arXiv preprint arXiv:2402.02941. [cs.CV].

# Fusion Strategies Comparison

**How to Combine CNN and Transformer Features**

| Strategy | Approach | Pros | Cons |
|---|---|---|---|
| Early Fusion | Input-level combination | Joint learning | Limited specialization |
| Late Fusion | Decision-level combination | Architectural independence | Missed interactions |
| Cross-Attention | Attention-based weighting | Adaptive fusion | High complexity |
| TransFuse BiFusion | Multi-modal attention | Sophisticated interaction | Implementation complexity |

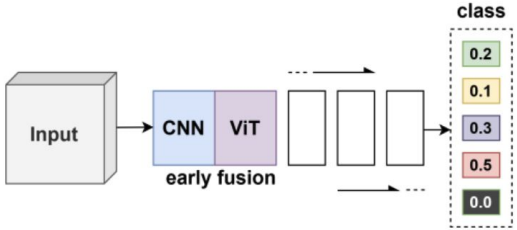



**Fig. 24** Early feature fusion

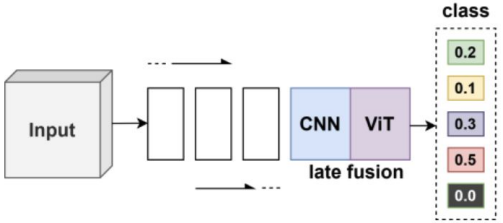



**Fig. 28** Late feature fusion

Yunusa et al. (2024)

# TransUNet (2022) - Sequential Hybrid Pioneer

**Architecture:**

- CNN encoder for initial feature extraction
- Many Transformer layers for global context
- Very strong Encoder
- Decoder combines local and global features

**Achievements:**

- "State-of-the-art results on multiple benchmarks"
- Proven hybrid concept viability

**Limitations:**

- Sequential dependency (CNN → Transformer)
- Information bottleneck between stages
- Limited parallel processing

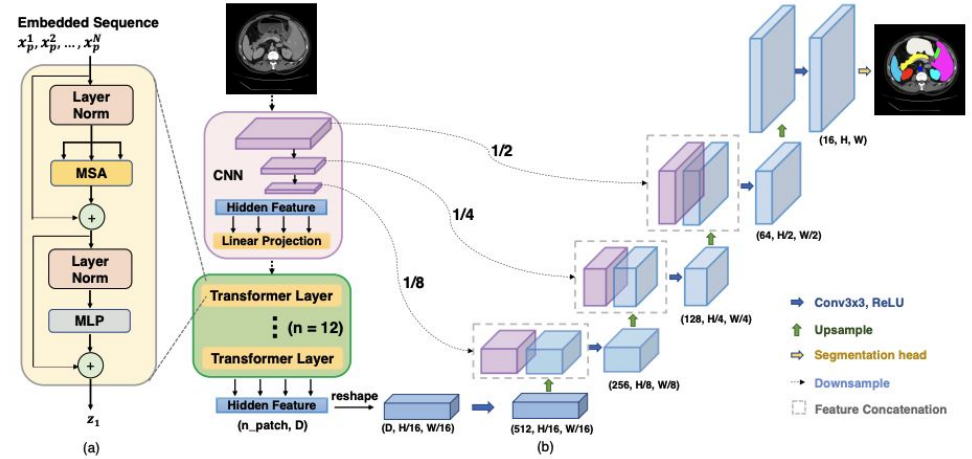



Fig. 1: Overview of the framework. (a) schematic of the Transformer layer; (b) architecture of the proposed TransUNet.

# TransFuse - Parallel Hybrid Innovation

**Architecture:**

- Parallel CNN (ResNet34) and Transformer (DeiT-S) branches
- BiFusion module with sophisticated attention mechanisms
- Adaptive feature selection per spatial location

**BiFusion Innovation:**

- Channel attention for Transformer features
- Spatial attention for CNN features
- Hadamard product for feature interaction

**Key Strength:** Superior results with parameter reduction

**Limitations:**

- **General medical imaging focus:** Not chest X-ray specific
- **Complex fusion:** High computational overhead for clinical deployment

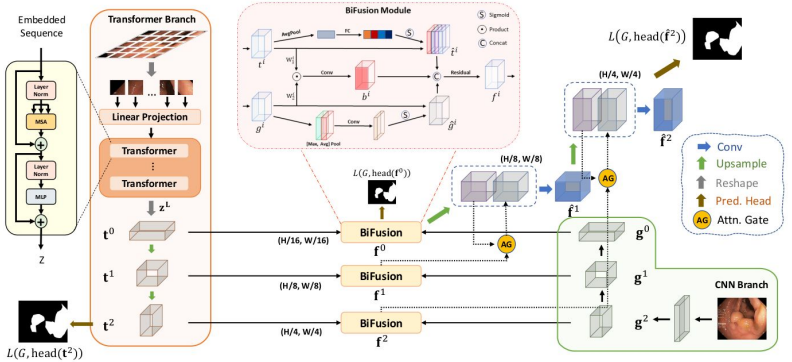
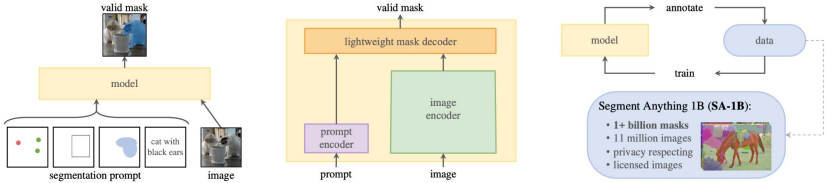



Fig. 1: Overview of TransFuse (best viewed in color), consists of two parallel branches - CNN (bottom right) and transformer (left). The two branches are fused with our proposed BiFusion module.

# Foundation Models - SAM Revolution (2023)

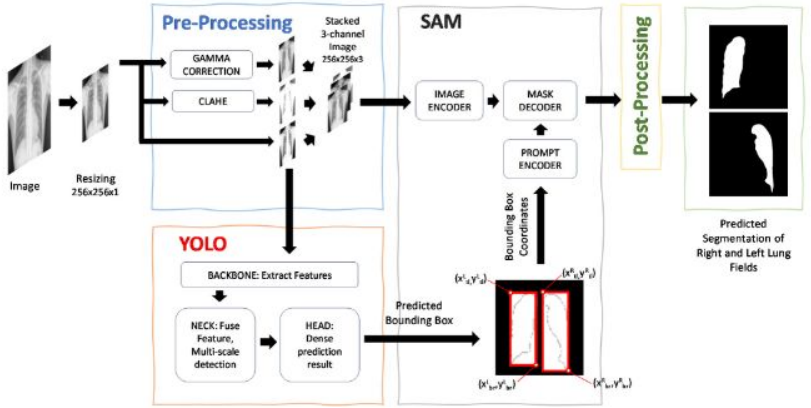
**SAM**

- **Foundation model:** Trained on 1 billion masks, 11 million images
- **Prompt-based segmentation:** Points, boxes, or masks as input
- **Zero-shot capability:** No task-specific training required




**YOLO-SAM Framework: (2024)**

- **Stage 1:** YOLO v5 detects lung fields → generates bounding boxes
- **Stage 2:** SAM segments lungs using bounding box prompts

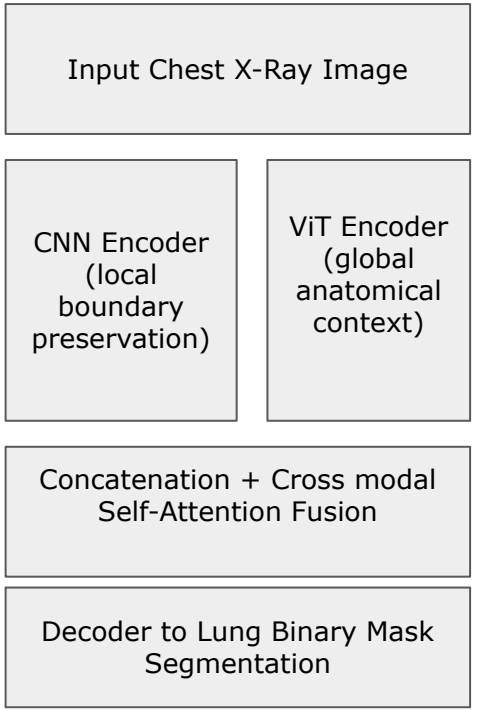



**Critical Dependencies:**

- **"Model effectiveness heavily depends on YOLO accuracy"**
- **Error propagation:** Detection failures cascade to segmentation
- **Two-stage complexity:** Requires coordination between models
- **Prompt engineering:** Manual bounding box optimization needed

# Current Research Focus & Gap

| Current Focus/Gap | Proposed approach |
|---|---|
| Most hybrid models target disease classification | Segmentation-specific hybrid design |
| General medical imaging applications, no Chest X-ray specifics | Chest X-ray domain optimization |
| Complex fusion attention mechanisms (BiFusion) | Efficient concatenation + multi head self-attention |
| External prompting dependencies (SAM) | End-to-end and single unified trainable architecture |
| Limited boundary preservation | CNN-Transformer harmonization |



# Research Objective & Questions

**Objective:**

- Develop hybrid CNN-Transformer architecture for chest X-ray lung segmentation accuracy

**Specific Research Questions:**

1. **Architecture Design:** How does parallel CNN-ViT feature processing with cross-modal attention compare to single-pathway and sequential hybrid approaches for lung segmentation?
2. **Cross-Modal Attention Fusion:** How does bidirectional cross-modal attention fusion balance local boundary precision and global anatomical coherence in lung segmentation?
3. **Performance Evaluation:** How does the proposed hybrid architecture perform relative to existing methods on standard lung segmentation benchmarks

# Research Contributions

**Primary Contribution:**

- A comprehensive hybrid CNN-ViT for chest X-ray lung segmentation

**Technical Innovations:**

- A  fusion strategy optimized for hybrid model of segmentation
- Chest X-ray specific architectural considerations
- Balance between performance and computational efficiency

**Clinical Impact:**

- Improved lung segmentation accuracy
- Foundation for automated chest X-ray analysis

**Research Community Value:**

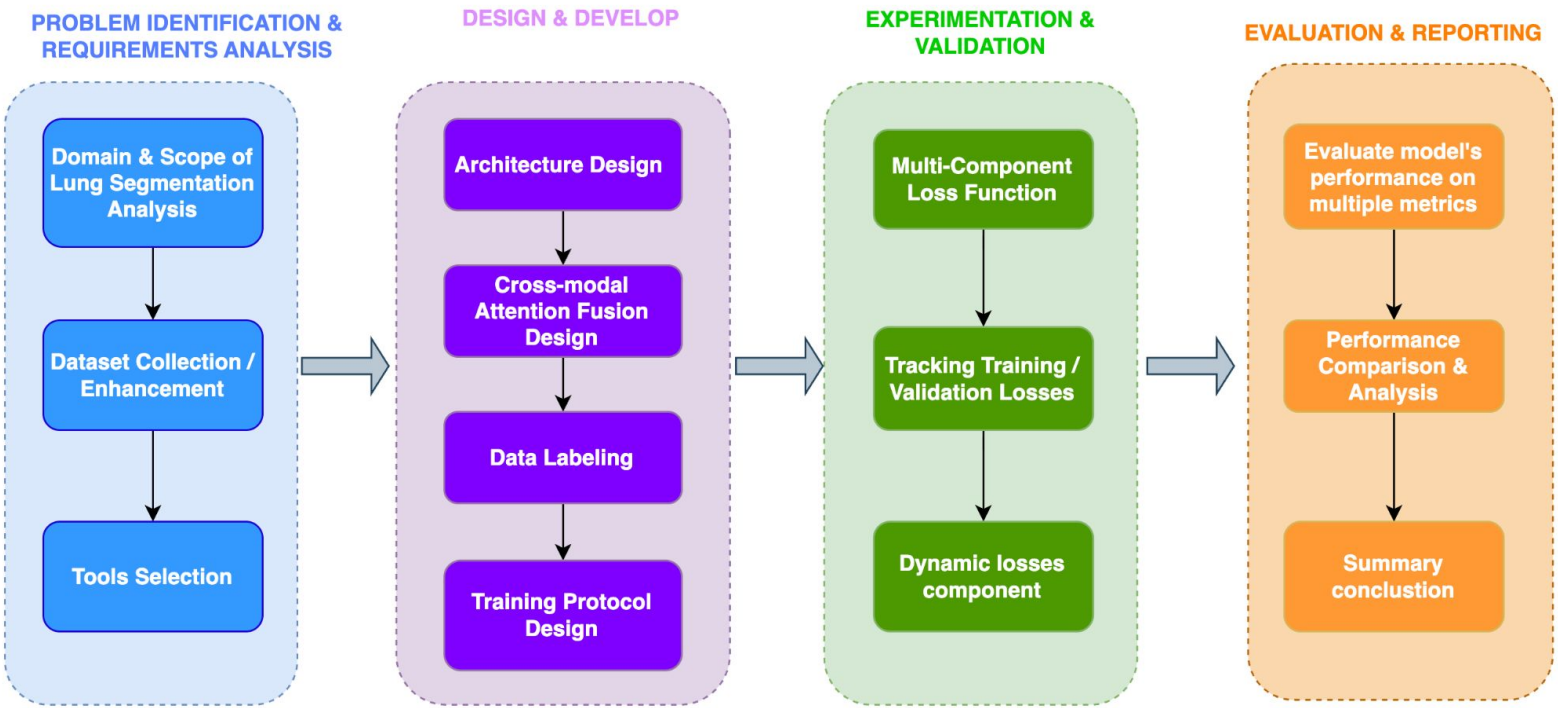
- Benchmark for chest X-ray segmentation
- Methodology applicable to other medical domains

# Methodology

# Methodology Flow

1. Overall Methodology process
2. Dataset description & Label process
3. Proposed Hybrid Model
4. Loss Function
5. Evaluation Metrics
6. Ablation studies

# Methodology Diagram



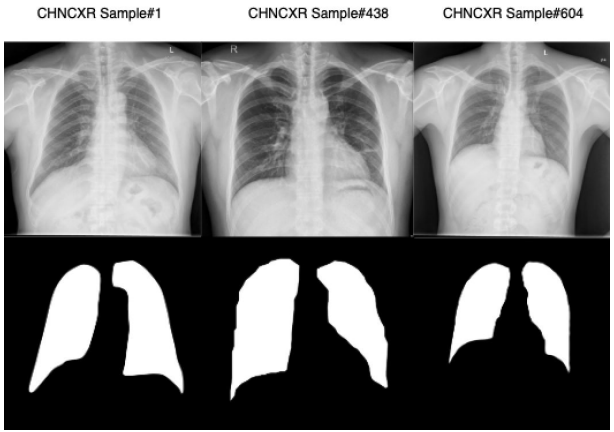
# Datasets

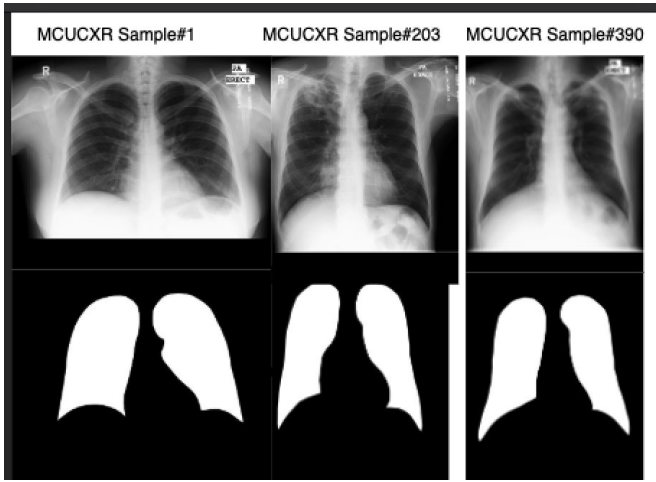
| Montgomery County Chest X-ray Dataset (MC) | |
|---|---|
| Total Images | **138** |
| Format | **PNG format, 12-bit grayscale images** |
| Resolution | **4,020×4,892** |
| Masks Available | **Yes by original research team** |
| Source | **Department of Health and Human Services, Montgomery County, Maryland, USA** |

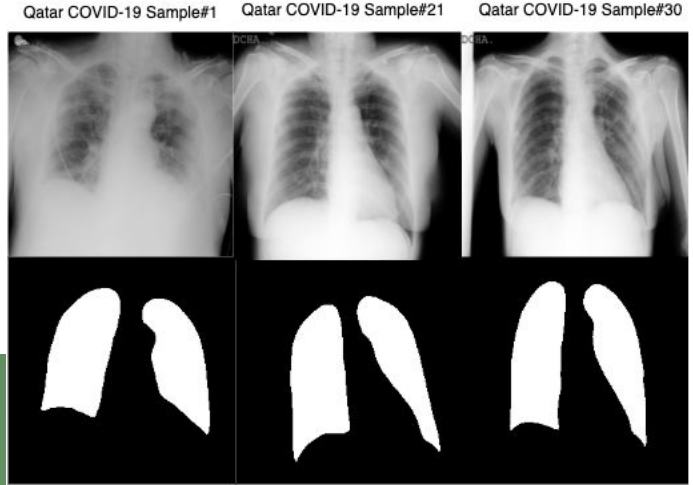
| Shenzhen (SH) Chest X-ray Dataset | |
|---|---|
| Total Images | **662** |
| Format | **PNG format, 12-bit grayscale images** |
| Resolution | **Variable, ~ 3000×3000** |
| Masks Available | **Manually by University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute", Kyiv, Ukraine** |
| Source | **Shenzhen No. 3 People's Hospital, China** |

| Qatar-COVID-19 Chest X-ray Dataset | |
|---|---|
| Total Images | **3616 COVID-19 cases<br>10192 normal cases** |
| Format | **PNG format, 12-bit grayscale images** |
| Resolution | **299x299** |
| Masks Available | **Based on standard dataset** |
| Source | **https://www.sciencedirect.com/science/article/pii/S001048252100113X?via%3Dihub#fig4** |

# Sample of pairs X-ray images vs ground truth masks



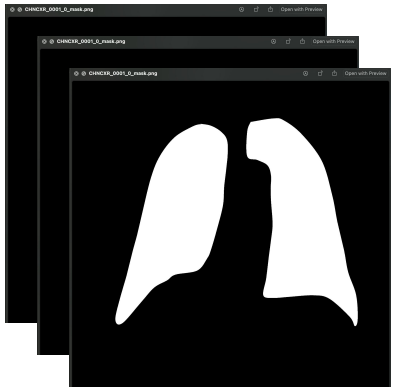
# Labeling Process

800 images = 138 MC + 662 SH



Original research team & University of Ukraine

704 Masks Ground truth

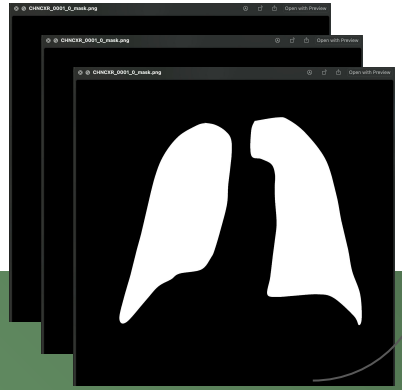


96 SH missing masks



Label studio

96 Masks Ground truth

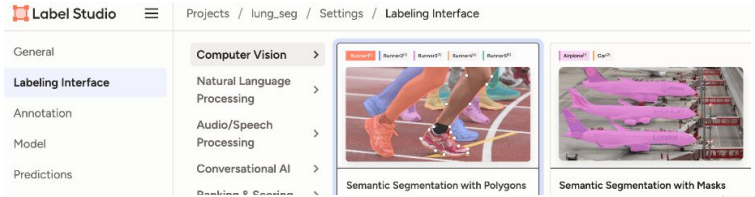


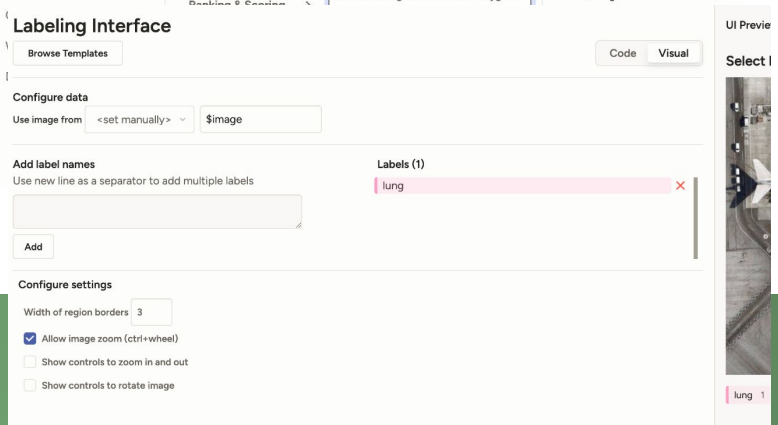
Full suite 800 chest x ray images with 800 masks ground truth

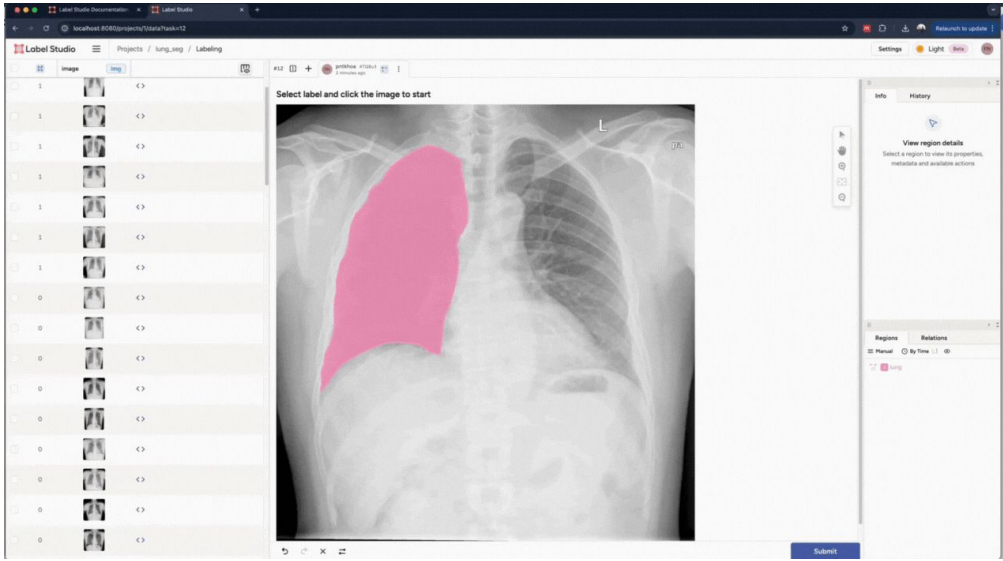
# Label Studio Process

## Label Process

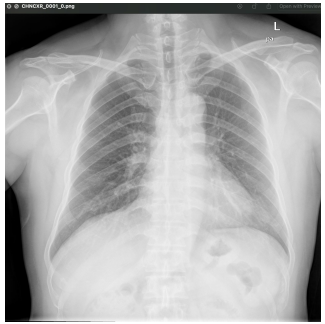
- Download Label Studio
- Create label project
- Choose Label interface semantic Segmentation with polygons
- Start labeling with polygons for both lungs
- Export Coco json format of labeled data
- Convert json coco format to binary masked



**Import image**



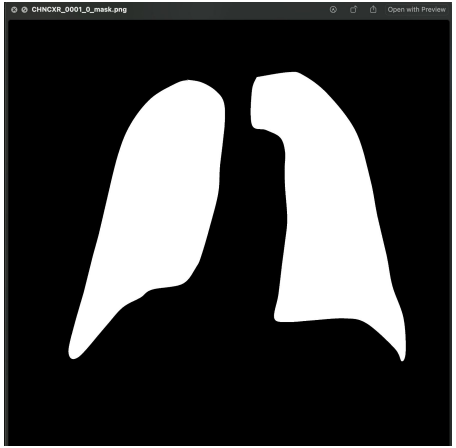
**Label by polygons**



**Export Json coco format**

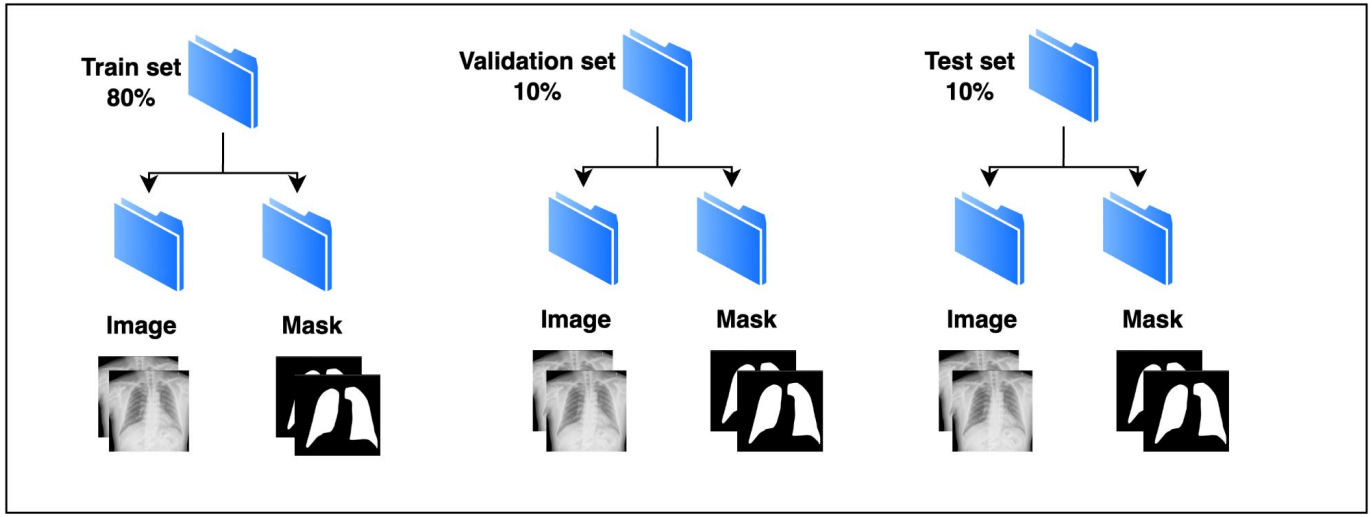

```
{
  "id": 3,
  "image_id": 1,
  "category_id": 0,
  "segmentation": [
    [
      1412.991133004926,
      369.2384236453202,..
      1543.823645320197,
      462.27487684729067
    ]
  ],
  "bbox": [
    630.9034482758622,
    369.2384236453202,
    930.3645320197046,
    2227.0600985221677
  ],
  "ignore": 0,
  "iscrowd": 0,
  "area": 1311038.1583625004
},
```


**Mask Creation based on segmentation info**

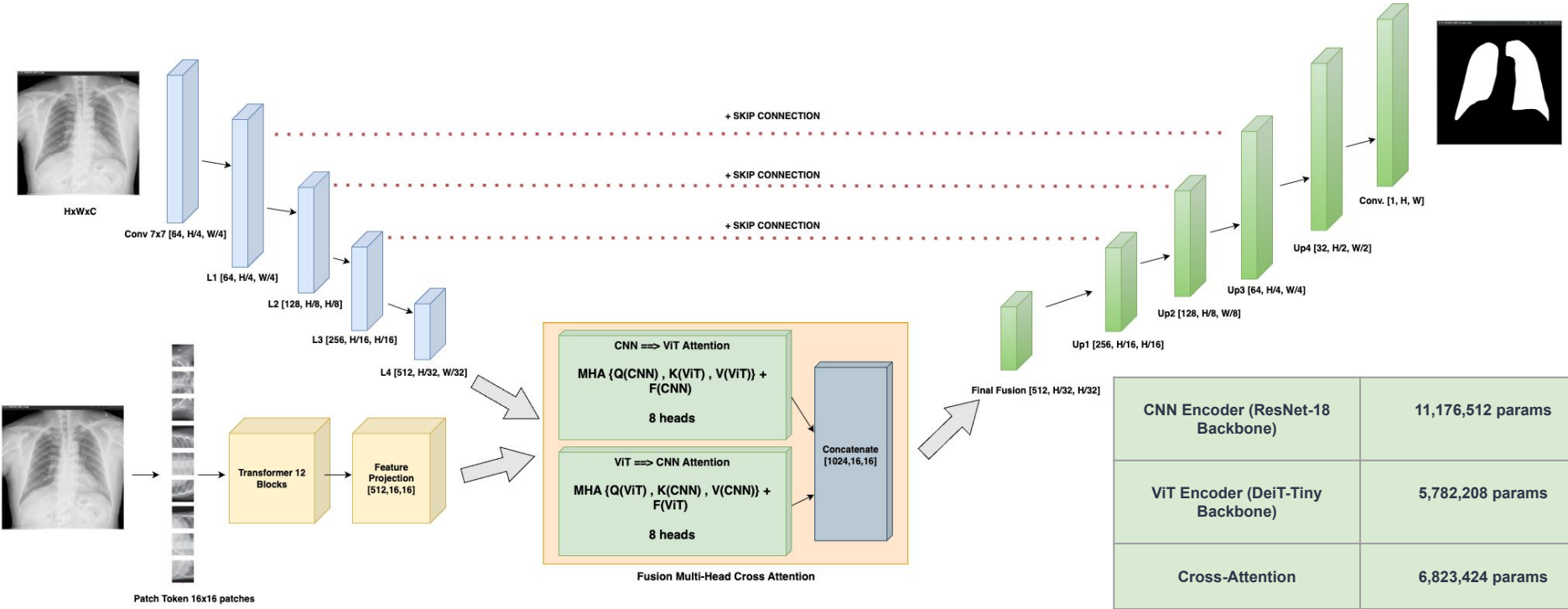



- Processing image: feb726eb-CHNCXR_0038_0.png -> CHNCXR_0038_0.png (ID: 3)
- Dimensions: 2744x3000
- Found annotation for category: lung
- Filled polygon with 58 points
- Found annotation for category: lung
- Filled polygon with 50 points
- Created mask: CHNCXR_0038_0_mask.png
- Mask contains values: [ 0 255]
- White pixels: 1820302/8232000 (22.1%)

# Images & Masks Folder Setup

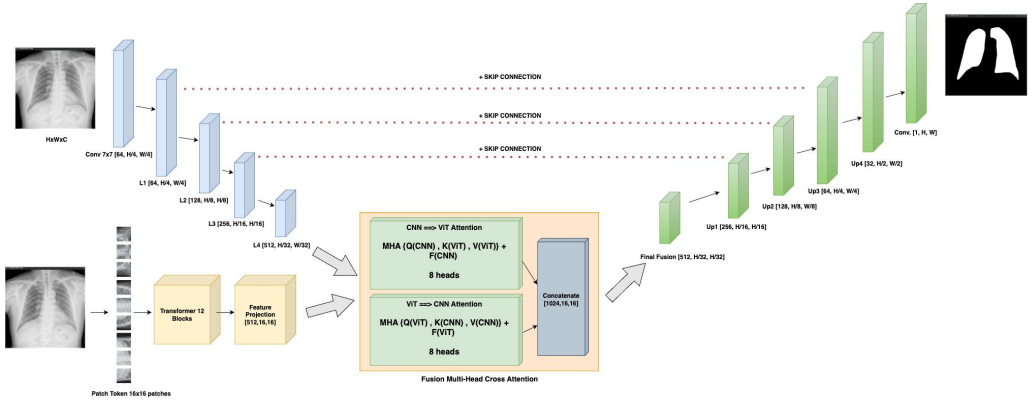


# Proposed Hybrid parallel CNN - Transformer Model for lung segmentation task




| | |
|---|---|
| CNN Encoder (ResNet-18 Backbone) | 11,176,512 params |
| ViT Encoder (DeiT-Tiny Backbone) | 5,782,208 params |
| Cross-Attention | 6,823,424 params |
| Decoder | 1,656,673 params |
| TOTAL PARAMETERS | 25,438,817 params |

# Model Architecture Description




a **parallel dual-pathway architecture** that combines the complementary strengths of Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) through a cross-modal attention fusion, specifically optimized for chest X-ray lung segmentation.

**CNN Branch Characteristics:** Layer-by-layer ResNet-18 breakdown with skip connections

- **Hierarchical Features**: Progressive spatial compression with channel expansion
- **Residual Learning**: Stable gradient flow through skip connections
- **Multi-Scale Outputs**: Features at 128×128, 64×64, 32×32, and 16×16 resolutions
- **Local Expertise**: Excels at edge detection, texture analysis, and spatial patterns

**ViT Branch**: DeiT-Tiny processing with patch embedding details

- **Global Receptive Field**: Every patch attends to every other patch from layer 1
- **Position Awareness**: Learned positional embeddings maintain spatial relationships
- **Semantic Understanding**: Self-attention captures long-range anatomical relationships
- **Patch-Based Processing**: 16×16 patches provide optimal granularity for chest X-rays

**Cross-Attention**:

- Q(CNN) → KV(ViT) Attention: "CNN learns from ViT's global understanding"
- Q(ViT) → KV(CNN) Attention: "ViT learns from CNN's local details"

**Decoder**: Progressive upsampling with skip connection integration

# Loss Function - Hypothesis

Medical image segmentation presents unique challenges that no single loss function can adequately address. The proposed approach employs a combined loss function that addresses multiple aspects of segmentation quality: class imbalance, boundary accuracy, overlap optimization, and training stability.

$$\mathcal{L}_{total} = \alpha\mathcal{L}_{BCE} + \beta\mathcal{L}_{Dice} + \gamma\mathcal{L}_{Focal} + \delta\mathcal{L}_{Boundary}$$

where $\alpha$, $\beta$, $\gamma$, and $\delta$ are dynamic weighting parameters that adapt throughout training phases.

$$\begin{bmatrix}\alpha \\ \beta \\ \gamma \\ \delta\end{bmatrix} = \begin{cases}[0.6, 0.2, 0.2, 0.0] & \text{if epoch} \leq 0.4 \times \text{total_epochs} \\ [0.3, 0.3, 0.2, 0.2] & \text{if } 0.4 < \text{epoch} \leq 0.8 \times \text{total_epochs} \\ [0.2, 0.3, 0.1, 0.4] & \text{if epoch} > 0.8 \times \text{total_epochs}\end{cases}$$

This adaptive weighting strategy optimizes learning progression by emphasizing stability in early training, balanced learning in mid-training, and boundary refinement in late training phases.

*In addition, also planning for* **unweighted 4 multi-losses configuration during training** *for evaluation and comparison the various protocol*

### BCE Loss

```
-[y*log(σ(x)) + (1-y)*log(1-σ(x))]
```

- Foundation loss function
- Stable gradients
- Leads early training (60%)
- Numerically stable with logits

### Focal Loss

```
FL(p_t) = -α(1-p_t)^γ * log(p_t)
```

- Handles class imbalance
- Focuses on hard examples
- Down-weights easy negatives
- α=0.25, γ=2.0 (defaults)

### Dice Loss

```
1 - 2|A∩B| / (|A|+|B|+ε)
```

- Direct IoU optimization
- Segmentation-specific
- Smooth and differentiable
- Handles imbalance naturally

### Boundary Loss

```
0.7*L_distance + 0.3*L_boundary
```

- Sharp boundary detection
- Distance transform weighting
- Morphological operations
- Spatial awareness

### Combined Loss

```
Σ w_i * L_i (dynamic weights)
```

- Adaptive training strategy
- Epoch-based weighting
- Best of all methods
- Individual tracking

### Dynamic Training Strategy

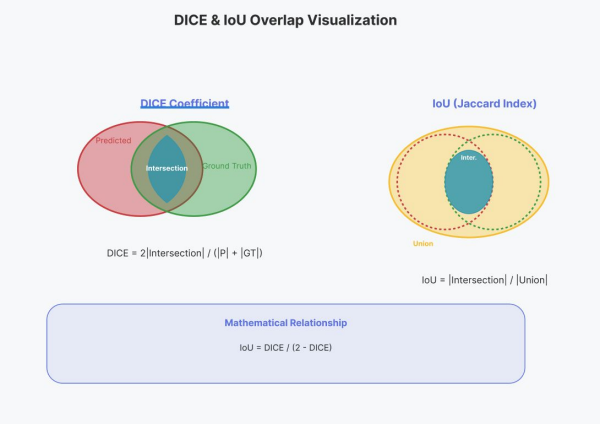
**Early Phase (0-40%)**
Focus on basic classification (BCE dominant)

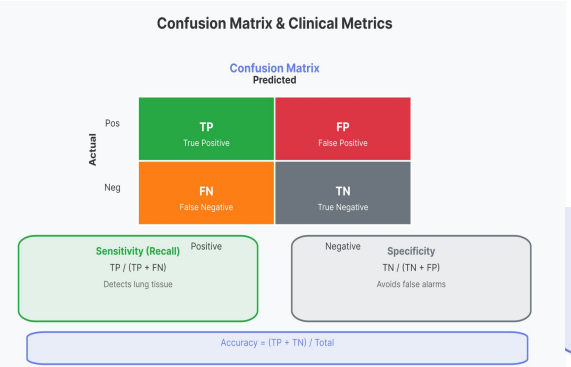
**Mid Phase (40-80%)**
Balanced approach (Dice emphasized)

**Late Phase (80-100%)**
Fine-tuning segmentation quality

BCE Loss · Dice Loss · Focal Loss · Boundary Loss

# Evaluation Metrics



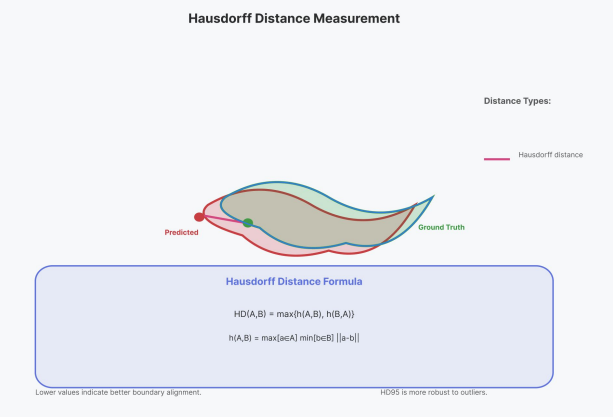




Precision = TP / (TP + FP)

How many detected boundaries are correct?

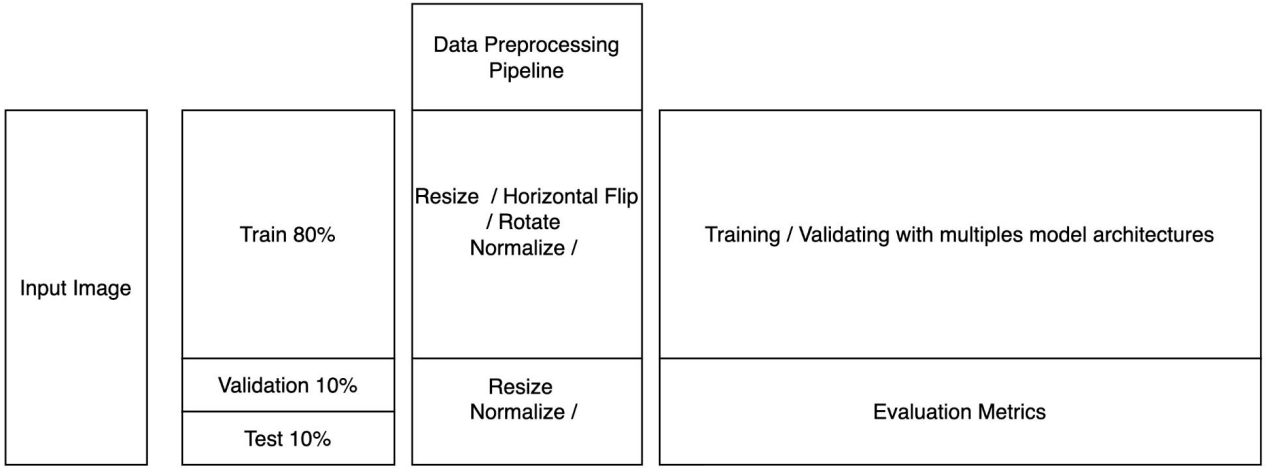
Recall = TP / (TP + FN)

How many true boundaries were detected?



# Experiment Results

# Overall Experiments Method




- Using Google Collab with A100 GPU
- Learning rate: 0.0005
- Epoch: 20 - 100
- Img H x Img W: 256x256

# 3 Training & Evaluation Scenarios

**01**

- MC+SHZ only (640 training, 80 validation, 80 test images)
- Training and testing on same dataset distribution
- Purpose: Establish baseline performance under controlled conditions

1. Hybrid model, Loss (BCE + Dice)
2. Hybrid model + 4 adaptive loss
3. CNN pathway only with 4 adaptive loss - no cross fusion
4. ViT pathway only with adaptive loss

**02**

- MC+SHZ training (640 training, 80 validation), Qatar-COVID-19 test set (3,616 never-seen images)
- Cross-dataset generalization evaluation
- Purpose: Evaluate generalization to different imaging protocols and populations

1. CNN pathway only with loss (BCE+Dice) - no cross fusion
2. ViT pathway only with loss (BCE+Dice) - no cross fusion
3. Hybrid model, Loss (BCE +Dice)
4. Hybrid model with 4 adaptive losses
5. Hybrid model with 4 unweight losses

**03**

- MC+SHZ+Normal (1,437 training, 280 validation), Qatar-COVID-19 test set (3,616 never-seen images)
- Increase number of training with diverse normal lung presentations
- Purpose: Assess impact of training data diversity on generalization

1. CNN pathway only with loss (BCE+Dice) - no cross fusion
2. ViT pathway only with loss (BCE+Dice) - no cross fusion
3. Hybrid model, Loss (BCE +Dice)
4. Hybrid model with 4 adaptive losses
5. Hybrid model with 4 unweight losses

# Scenario #1 - Baseline Performance

**In-Domain Performance (MC+SHZ Dataset): All models achieved strong baseline with minimal differences**

**Table 4.1**

*Performance comparison for models trained and tested on the MC+SHZ dataset. Best results are highlighted in bold.*

| Model Configuration | Acc (%) | Dice (%) | IoU (%) | Sens (%) | Spec (%) | HD | HD95 |
|---|---|---|---|---|---|---|---|
| Hybrid (BCE+Dice Losses) | 98.32 ± 0.31 | 96.49 ± 0.75 | 93.22 ± 1.40 | 96.11 ± 0.71 | 99.02 ± 0.23 | 19.68 ± 3.38 | 8.70 ± 1.83 |
| Hybrid (Adaptive Losses) | **98.38 ± 0.16** | **96.65 ± 0.54** | **93.52 ± 1.01** | 96.13 ± 0.35 | **99.11 ± 0.16** | **19.17 ± 2.29** | **8.23 ± 1.23** |
| CNN only (Adaptive Losses) | 98.33 ± 0.17 | 96.52 ± 0.57 | 93.28 ± 1.05 | 95.70 ± 0.42 | **99.17 ± 0.16** | 19.53 ± 2.82 | 9.65 ± 2.55 |
| ViT only (Adaptive Losses) | 98.35 ± 0.21 | 96.60 ± 0.65 | 93.43 ± 1.21 | **96.50 ± 0.37** | 98.95 ± 0.21 | 19.40 ± 2.04 | 8.55 ± 1.70 |

**Key Findings**

- Tight Clustering: All models 96.5-96.7% Dice (0.16% range) - minimal differences on matched data
- Hybrid Adaptive Best: Highest Dice, lowest HD95, most stable
- ViT Competitive: 96.60% Dice, challenges CNN superiority for segmentation
- CNN Conservative: Highest specificity (99.17%), lowest sensitivity (95.70%)
- Accuracy Insufficient: 98.32-98.38% range too narrow for discrimination
- Hybrid: balanced choice; ViT: global consistency; CNN: high specificity

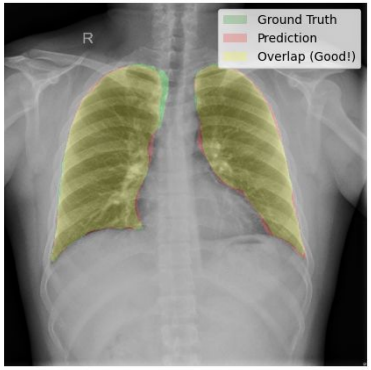
## Scenario 1: CNN only

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



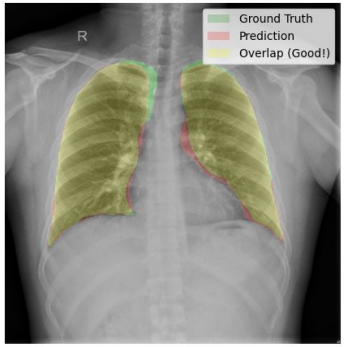

## Scenario 1: Hybrid BCE + Dice Loss only

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



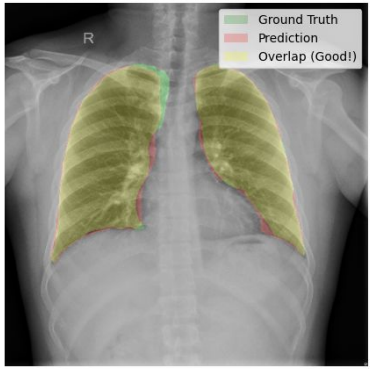

## Scenario 1: ViT only

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



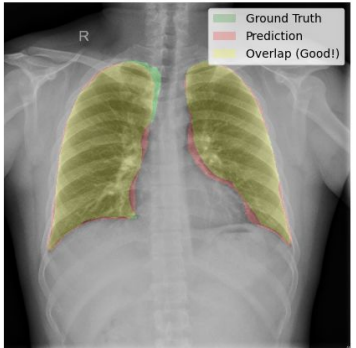

## Scenario 1: Hybrid adaptive losses

Sample 1 - Combined Overlay Analysis

Original Image



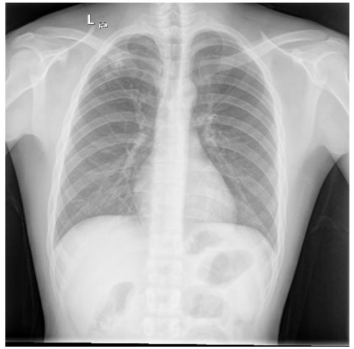
Original + GT(Green) + Pred(Red)



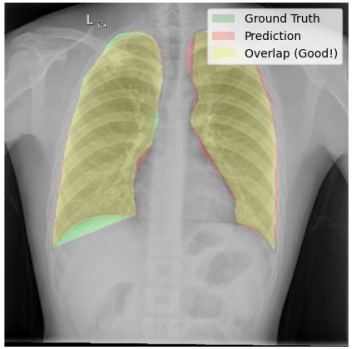
## Scenario 1: CNN only

Sample 2 - Combined Overlay Analysis

Original Image



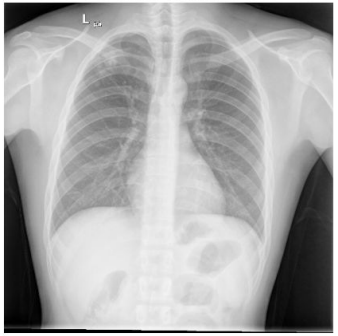
Original + GT(Green) + Pred(Red)



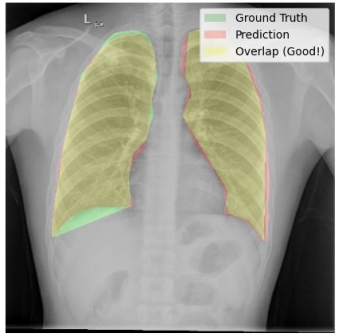

## Scenario 1: Hybrid BCE + Dice Loss only

Sample 2 - Combined Overlay Analysis

Original Image



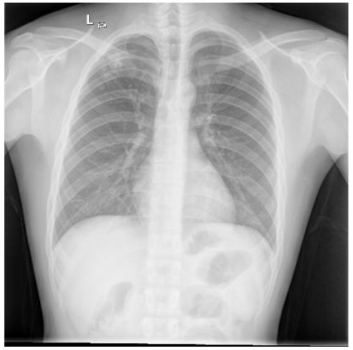
Original + GT(Green) + Pred(Red)



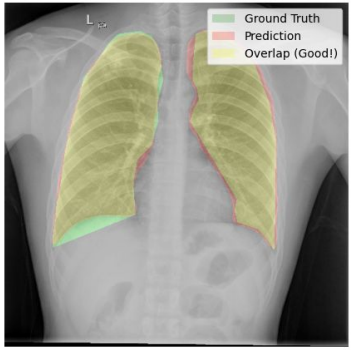

## Scenario 1: ViT only

Sample 2 - Combined Overlay Analysis

Original Image



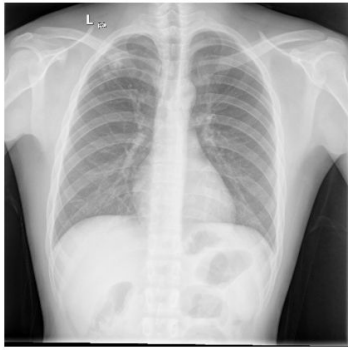
Original + GT(Green) + Pred(Red)



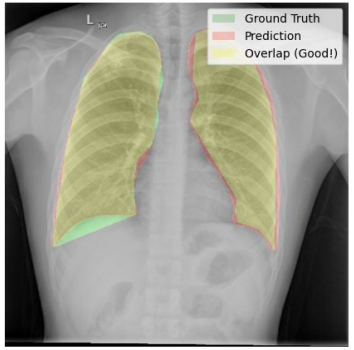

## Scenario 1: Hybrid adaptive losses

Sample 2 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



# Scenario #2 DOMAIN SHIFT (Cross-Dataset)

**Table 4.2**

*Cross-dataset generalization performance. Models were trained on MC+SHZ and tested on the unseen Qatar-COVID-19 dataset.*

| Model | Acc (%) | Dice (%) | IoU (%) | Sens (%) | Spec (%) | Prec (%) | HD | HD95 |
|---|---|---|---|---|---|---|---|---|
| CNN-Only | 96.32 ± 1.66 | 92.27 ± 3.37 | 85.82 ± 5.70 | 89.67 ± 6.54 | 98.60 ± 0.35 | 95.34 ± 1.18 | 50.84 ± 16.39 | 26.69 ± 12.14 |
| ViT-Only | 97.17 ± 1.25 | 94.14 ± 2.35 | 89.02 ± 4.13 | 92.17 ± 4.45 | 98.87 ± 0.37 | 96.33 ± 1.12 | 40.60 ± 13.76 | 19.74 ± 9.79 |
| Hybrid (BCE+Dice Losses) | **97.38 ± 1.13** | **94.56 ± 2.15** | **89.75 ± 3.81** | **92.52 ± 4.19** | 99.02 ± 0.27 | **96.80 ± 0.87** | **34.05 ± 11.44** | **16.87 ± 8.09** |
| Hybrid (Adaptive Losses) | 97.27 ± 1.06 | 94.34 ± 1.97 | 89.35 ± 3.51 | 92.08 ± 3.90 | **99.03 ± 0.20** | **96.80 ± 0.80** | 43.18 ± 10.38 | 18.43 ± 7.00 |
| Hybrid (Unweighted Losses) | 97.14 ± 1.11 | 94.09 ± 2.08 | 88.91 ± 3.67 | 92.46 ± 4.28 | 98.74 ± 0.24 | 95.91 ± 0.96 | 33.17 ± 9.88 | 17.04 ± 7.24 |

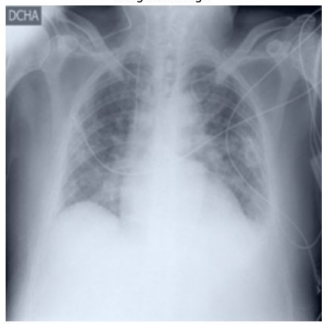
**Key Findings**

- Degradation: Accuracy -1% | Dice -2% | IoU -4% | HD95 +105% (doubled)
- Architecture Shift: Dice Score ViT now outperforms CNN (94.14% vs 92.27%) - ViT generalizes better
- CNN Collapse: Dice Score - Worst drop (-4.25%), highest variance (±3.37%), catastrophic boundaries
- Variance Dramatically increase: unstable performance
- Sensitivity Drop: 4-7 points lost; Specificity maintained - lung appearance varies, background stable
- Hybrid BCE+Dice generally outperformed Adaptive configuration contrary to Scenario 01 patterns, suggesting loss function optimized for one domain may not transfer optimally.

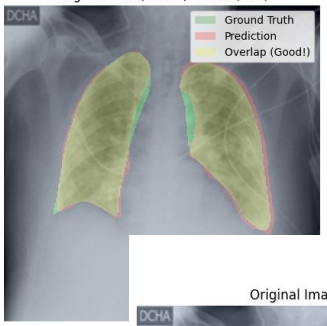
## Scenario 2: CNN Only

Sample 4 - Combined Overlay Analysis

Original Image



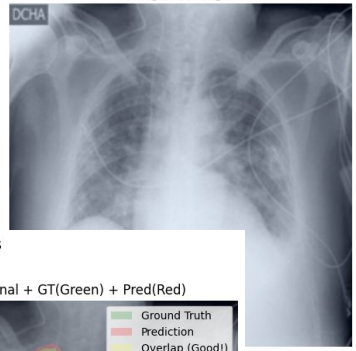

Original + GT(Green) + Pred(Red)



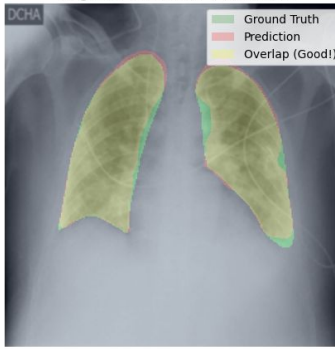

## Scenario 2: ViT Only

Sample 4 - Combined Overlay Analysis

Original Image



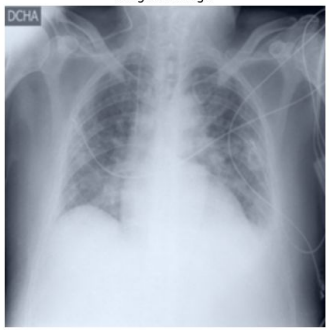

Original + GT(Green) + Pred(Red)



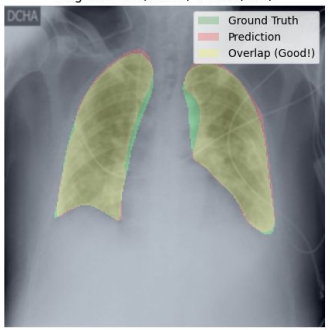

## Scenario 2: Hybrid BCE + Dice Losses

Sample 4 - Combined Overlay Analysis

Original Image



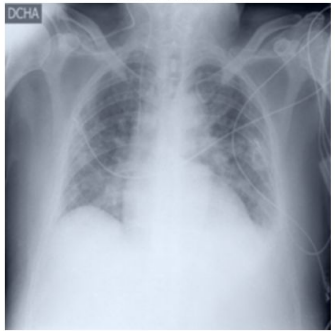

Original + GT(Green) + Pred(Red)



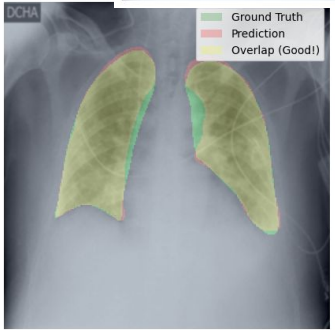

## Scenario 2: Hybrid Adaptive Losses

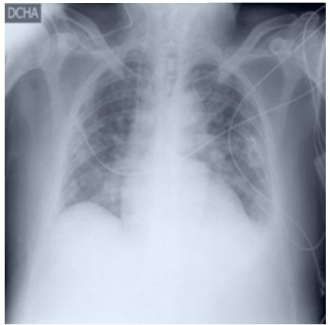
Sample 4 - Combined Overlay Analysis

Original Image




Original

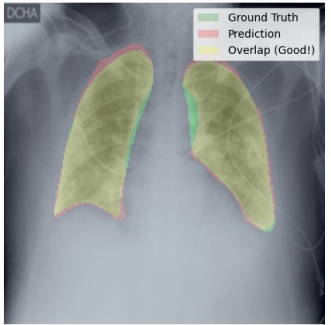



## Scenario 2: Hybrid Unweighted Losses

Sample 4 - Combined Overlay Analysis

Original + GT(Green) + Pred(Red)



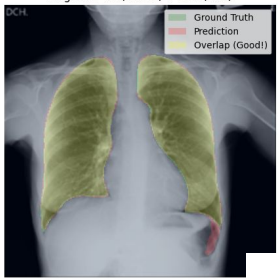
## Scenario 2: CNN Only

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



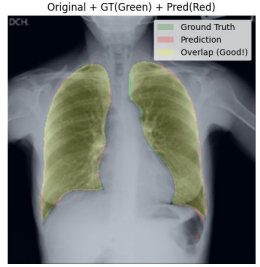

## Scenario 2: ViT Only

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



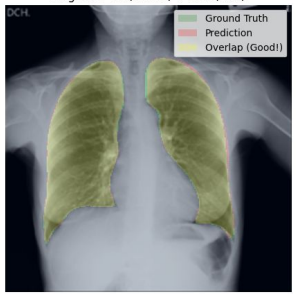

## Scenario 2: Hybrid BCE + Dice Losses

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



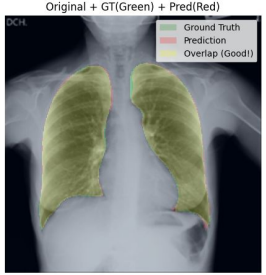

## Scenario 2: Hybrid Adaptive Losses

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



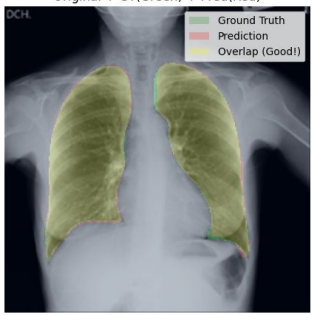

## Scenario 2: Hybrid Unweighted Losses

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



# Scenario #3 Enhanced Training

**Table 4.3**

*Performance on the Qatar-COVID-19 test set after enhanced training with MC+SHZ and additional normal COVID-19 images.*

| Model | Acc (%) | Dice (%) | IoU (%) | Sens (%) | Spec (%) | Prec (%) | HD | HD95 |
|---|---|---|---|---|---|---|---|---|
| ViT Only | 98.01 ± 0.86 | 95.96 ± 1.56 | 92.28 ± 2.83 | 95.58 ± 2.63 | 98.85 ± 0.42 | 96.39 ± 1.36 | 27.32 ± 10.84 | 13.27 ± 7.00 |
| CNN Only | 98.08 ± 1.02 | 96.06 ± 1.92 | 92.48 ± 3.49 | 94.66 ± 3.73 | 99.25 ± 0.20 | 97.59 ± 0.69 | 26.40 ± 11.23 | 12.22 ± 6.74 |
| Hybrid (BCE+Dice Losses) | 98.29 ± 0.86 | 96.52 ± 1.58 | 93.31 ± 2.90 | 95.21 ± 3.11 | **99.36 ± 0.16** | **97.92 ± 0.56** | 20.85 ± 9.45 | 10.04 ± 6.04 |
| Hybrid (Adaptive Losses) | 98.29 ± 0.84 | 96.51 ± 1.53 | 93.31 ± 2.81 | 95.55 ± 3.05 | 99.24 ± 0.17 | 97.56 ± 0.64 | 20.59 ± 9.42 | 9.90 ± 6.07 |
| Hybrid (Unweighted Losses) | **98.38 ± 0.73** | **96.71 ± 1.32** | **93.67 ± 2.43** | **96.42 ± 2.35** | 99.06 ± 0.32 | 97.05 ± 0.97 | **20.28 ± 9.29** | **9.81 ± 5.92** |

**Key Findings**

- All architectures benefit from enhanced diverse training data
- Recovery was most dramatic for IoU (6–7 points), sensitivity (3–5 points), and boundary metrics (halving HD95), while accuracy and specificity improved modestly since they were already near-ceiling in Scenario 02.
- Variance Reduced 40%: ±2.15% → ±1.32% - more consistent, predictable performance (Dice Score)
- CNN-only benefits most from diverse training data
- Hybrid Unweighted achieved best overall performance across multiple metrics (Dice, IoU, Sensitivity, HD), despite adaptive loss weighting showing advantages in Scenario 01

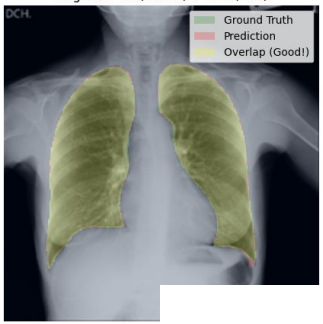
## Scenario 3: CNN Only

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



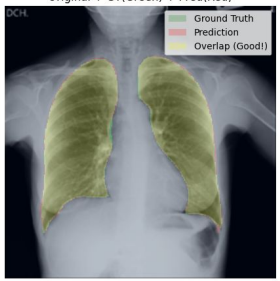

## Scenario 3: ViT Only

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



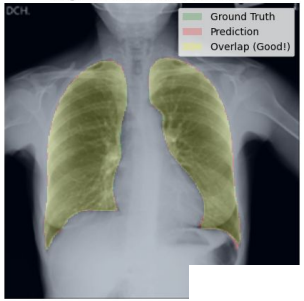

## Scenario 3: Hybrid BCE + Dice Losses

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



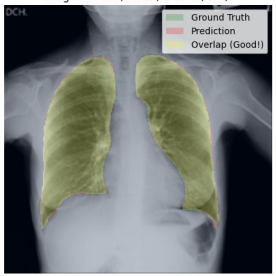

## Scenario 3: Hybrid Adaptive Losses

Sample 1 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



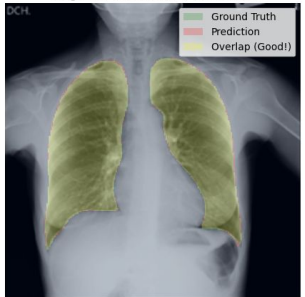

## Scenario 3: Hybrid Unweighted Losses

Sample 1 - Combined Overlay Analysis

Original Image



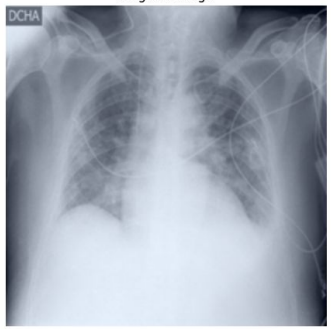
Original + GT(Green) + Pred(Red)



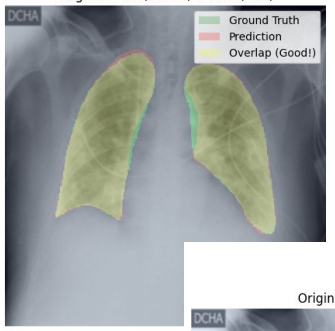
## Scenario 3: CNN Only

Sample 4 - Combined Overlay Analysis

Original Image



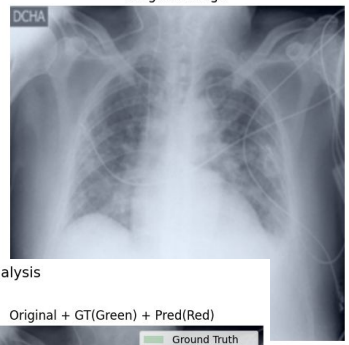
Original + GT(Green) + Pred(Red)



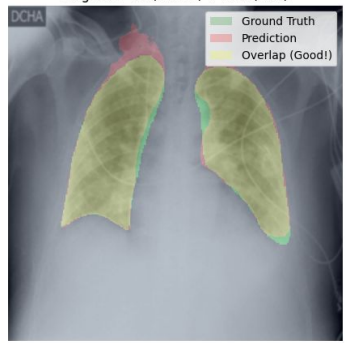

## Scenario 3: ViT Only

Sample 4 - Combined Overlay Analysis

Original Image



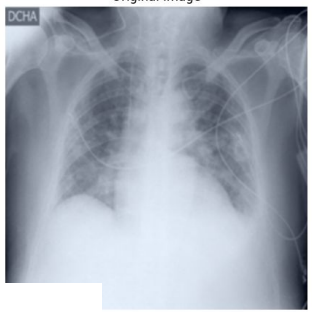
Original + GT(Green) + Pred(Red)



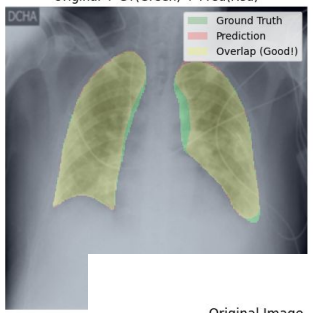

## Scenario 3: Hybrid BCE + Dice Losses

Sample 4 - Combined Overlay Analysis

Original Image



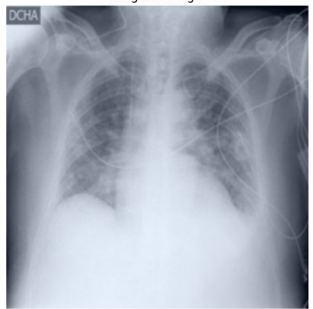
Original + GT(Green) + Pred(Red)



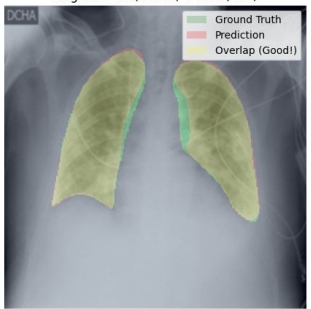

## Scenario 3: Hybrid Adaptive Losses

Sample 4 - Combined Overlay Analysis

Original Image



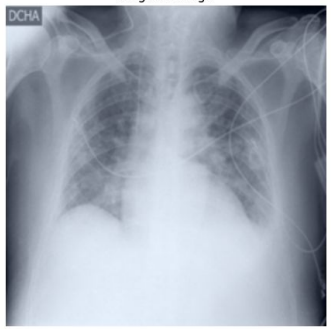
Original + GT(Green) + Pred(Red)



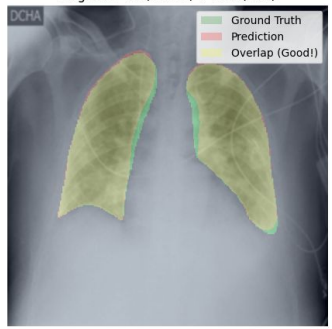

## Scenario 3: Hybrid Unweighted Losses

Sample 4 - Combined Overlay Analysis

Original Image



Original + GT(Green) + Pred(Red)



# Previous Work Benchmark

**Performance Highlights (M+S Dataset)**

- **Our Method**: 96.65% DSC, 8.23 HD95, 93.52% IoU - Competitive across all standard metrics
- Superior boundary precision compared to recent methods

1. **vs. TransUNet (2021)**
   - +0.20% Dice improvement
   - 23% better boundary precision (HD95: 8.23 vs 10.75 pixels)
2. **vs. U-Net++ and Swin-Unet**
   - HD95 improvements: 30-32% better
   - Clear advantages in both overlap and boundary localization
3. **vs. FAT-Net (2025)**
   - +0.55% Dice improvement (96.65% vs 96.10%)
   - Superior boundary metrics (HD95: 8.23 pixels)

- Modest overlap improvements translate to **substantial boundary precision gains**
- 23-32% better boundary localization = more accurate lung volume measurements
- Critical for detecting pathology near lung boundaries
- Our hybrid CNN-ViT approach achieves **competitive overlap** with **high performance boundary precision**

**Table 4.4**

*Comparison with state-of-the-art methods. Dataset abbreviations: M=MC, S=SHZ, C=COVID19, P=Private, Q=QaTa-COV19. Results for FAT-Net are from Khorasani and Mofrad Khorasani and Mofrad (2025).*

| Research | Method | Dataset | Acc (%) | Dice (%) | Year |
|---|---|---|---|---|---|
| | VAE | M+S+C | 88.15 | - | 2020 |
| | U-Net | M+S | 91.47 | - | 2022 |
| | ConvMixer | M+S | 97.52 | - | 2023 |
| | VAE-AL-UNet | M+S+P | - | 95.00 | 2023 |
| | DECTNet | Q | 97.50 | 87.99 | 2024 |
| | Swin U-Net | M+S | 97.80 | 95.71 | 2024 |
| Khorasani and Mofrad (2025) | FAT-Net | M+S | 98.12 | 96.10 | 2025 |
| **Ours** | **Hybrid + Adaptive Losses** | **M+S** | **98.38** | **96.65** | **2025** |
| **Ours** | **Hybrid + Unweighted Losses** | **M+S+Q** | **98.38** | **96.71** | **2025** |

**Table 4.5**

*Performance comparison on the Shenzhen dataset. Benchmark results for U-Net, UNet++, TransUNet, and Swin-Unet are adapted from Yao et al. Yao et al. (2023).*

| Method | DSC (%) | HD95 | IoU (%) | Acc (%) | Prec (%) | Recall (%) |
|---|---|---|---|---|---|---|
| U-Net | 95.32 | 14.23 | 91.24 | 97.69 | 96.13 | 94.78 |
| UNet++ | 95.83 | 11.75 | 92.15 | 97.95 | 97.31 | 94.62 |
| TransUNet | 96.45 | 10.75 | 93.25 | 98.16 | 97.36 | 95.72 |
| Swin-Unet | 95.71 | 12.10 | 91.88 | 97.80 | 96.81 | 94.79 |
| **Ours** | **96.65** | **8.23** | **93.52** | **98.38** | **97.17** | **96.13** |

# Ablation Study - Architecture components

- Architectural fusion benefits scale with domain shift severity—minimal under matched distributions but substantial under cross-dataset conditions
- CNNs and ViTs exhibit complementary robustness profiles—CNNs excel with sufficient training diversity while ViTs demonstrate superior generalization under data scarcity or domain shift
- The Full Hybrid (CNN+ViT) model consistently outperforms single-architecture models across all three scenarios, demonstrating the value of combining convolutional and transformer components.

**Table 4.6**

*Performance comparison of the full hybrid model and its architectural ablations (CNN-Only, ViT-Only) across three experimental scenarios. The 'Full Hybrid' model serves as the baseline, with ΔDice and ΔHD95 metrics calculated relative to its performance in each scenario. Scenario 1 (Standard Training) tests on the MC+SHZ dataset. Scenarios 2 (Cross-Dataset Generalization) and 3 (Expanded Training) test on the unseen Qatar-COVID-19 dataset.*

| Configuration | Test Dataset | Training Data | Dice (%) | ΔDice vs Hybrid | HD95 | ΔHD95 vs Hybrid |
|---|---|---|---|---|---|---|
| ***Scenario 1: Standard Training (640 images)*** | | | | | | |
| **Full Hybrid** | **MC+SHZ** | **640** | **96.65 ± 0.54** | **–** | **8.23 ± 1.23** | **–** |
| w/o ViT (CNN-Only) | MC+SHZ | 640 | 96.52 ± 0.57 | −0.13% | 9.65 ± 2.55 | +1.42 |
| w/o CNN (ViT-Only) | MC+SHZ | 640 | 96.60 ± 0.65 | −0.05% | 8.55 ± 1.70 | +0.32 |
| ***Scenario 2: Cross-Dataset Generalization (640 training → COVID-19 testing)*** | | | | | | |
| **Full Hybrid** | **COVID-19** | **640** | **94.56 ± 2.15** | **–** | **16.87 ± 8.09** | **–** |
| w/o ViT (CNN-Only) | COVID-19 | 640 | 92.27 ± 3.37 | −2.29% | 26.69 ± 12.14 | +9.82 |
| w/o CNN (ViT-Only) | COVID-19 | 640 | 94.14 ± 2.35 | −0.42% | 19.74 ± 9.79 | +2.87 |
| ***Scenario 3: Expanded Training (1,437 images → COVID-19 testing)*** | | | | | | |
| **Full Hybrid** | **COVID-19** | **1,437** | **96.52 ± 1.58** | **–** | **10.04 ± 6.04** | **–** |
| w/o ViT (CNN-Only) | COVID-19 | 1,437 | 96.06 ± 1.92 | −0.46% | 12.22 ± 6.74 | +2.18 |
| w/o CNN (ViT-Only) | COVID-19 | 1,437 | 95.96 ± 1.56 | −0.56% | 13.27 ± 7.00 | +3.23 |

# Ablation Study - Loss components

**Optimal loss strategy depends on data regime—adaptive weighting for limited matched data, simple BCE+Dice for domain shift, and multi-term fixed weighting for abundant diverse training.**

**Unweighted 4 multi-losses configuration during training** is the optimal choice for lung segmentation because:

1. **Superior accuracy** across primary metrics (Dice, HD95)
2. **Enhanced stability** with lower variability in Dice scores
3. **Simpler implementation** - no hyperparameter tuning required for loss weights

**Table 4.3**

*Performance on the Qatar-COVID-19 test set after enhanced training with MC+SHZ and additional normal COVID-19 images.*

| Model | Acc (%) | Dice (%) | IoU (%) | Sens (%) | Spec (%) | Prec (%) | HD | HD95 |
|---|---|---|---|---|---|---|---|---|
| ViT Only | 98.01 ± 0.86 | 95.96 ± 1.56 | 92.28 ± 2.83 | 95.58 ± 2.63 | 98.85 ± 0.42 | 96.39 ± 1.36 | 27.32 ± 10.84 | 13.27 ± 7.00 |
| CNN Only | 98.08 ± 1.02 | 96.06 ± 1.92 | 92.48 ± 3.49 | 94.66 ± 3.73 | 99.25 ± 0.20 | 97.59 ± 0.69 | 26.40 ± 11.23 | 12.22 ± 6.74 |
| Hybrid (BCE+Dice Losses) | 98.29 ± 0.86 | 96.52 ± 1.58 | 93.31 ± 2.90 | 95.21 ± 3.11 | **99.36 ± 0.16** | **97.92 ± 0.56** | 20.85 ± 9.45 | 10.04 ± 6.04 |
| Hybrid (Adaptive Losses) | 98.29 ± 0.84 | 96.51 ± 1.53 | 93.31 ± 2.81 | 95.55 ± 3.05 | 99.24 ± 0.17 | 97.56 ± 0.64 | 20.59 ± 9.42 | 9.90 ± 6.07 |
| Hybrid (Unweighted Losses) | **98.38 ± 0.73** | **96.71 ± 1.32** | **93.67 ± 2.43** | **96.42 ± 2.35** | 99.06 ± 0.32 | 97.05 ± 0.97 | **20.28 ± 9.29** | **9.81 ± 5.92** |

**Table 4.7**

*Performance comparison of different loss configurations for the hybrid model in Scenario 3. All Δ metrics are relative to the 'Unweighted Losses' baseline. Bold values indicate improvements over baseline; regular font indicates degradation. Note: For HD and HD95, negative Δ values represent improvements (lower is better).*

| Loss Configuration | ΔAcc | ΔDice | ΔIoU | ΔSens | ΔSpec | ΔPrec | ΔHD† | ΔHD95† |
|---|---|---|---|---|---|---|---|---|
| **Unweighted Losses (Baseline)** | – | – | – | – | – | – | – | – |
| BCE+Dice (Weighted) | −0.09% | −0.19% | −0.36% | −1.21% | **+0.30%** | **+0.87%** | +0.57 | +0.23 |
| Adaptive Losses | −0.09% | −0.20% | −0.36% | −0.87% | **+0.18%** | **+0.51%** | +0.31 | +0.09 |

† For Hausdorff Distance metrics, negative values indicate improvement (shorter distance is better).

# Parameters Comparison

Key Findings: Parameters vs. Performance

ViT-Only (6M params)

- Most parameter-efficient option
- Dice: 95.96% | HD95: 13.27 pixels
- Good baseline performance with minimal computational cost

CNN-Only (12M params)

- 2× parameters vs. ViT
- Dice: 96.06% | HD95: 12.22 pixels
- Marginal improvements over ViT-Only

Hybrid Models (25M params)

- 4× parameters vs. ViT, 2× vs. CNN
- Dice: 96.51-96.71% | HD95: 9.81-10.04 pixels
- **Key insight: 18-26% reduction in boundary errors**

**Table 4.8**

*Model Architecture Performance and Efficiency Trade-offs*

| Model | Params | Dice (%) | HD95 |
|---|---|---|---|
| CNN-Only | 12M | 96.06 ± 1.92 | 12.22 ± 6.74 |
| ViT-Only | **6M*** | 95.96 ± 1.56 | 13.27 ± 7.00 |
| Hybrid (BCE+Dice Losses) | 25M | 96.52 ± 1.58 | 10.04 ± 6.04 |
| Hybrid (Adaptive) | 25M | 96.51 ± 1.53 | 9.90 ± 6.07 |
| Hybrid (Unweighted Losses) | 25M | **96.71 ± 1.32** | **9.81 ± 5.92** |

# Conclusion

**Objective:** To develop a hybrid CNN-Transformer architecture for accurate and robust lung segmentation.

- **Key Achievement:** Achieved a top Dice score of **96.71%** (HD95 = 9.81) and demonstrated superior boundary precision compared to SOTA methods
- **Architectural Finding (RQ1):** A **parallel CNN-ViT pathway** design is more robust to data shifts and unseen data than single-pathway models.
- **Fusion Finding (RQ2): Bidirectional cross-modal attention** was proven to be highly effective, balancing the CNN's local detail with the ViT's global coherence.
- **Performance Evaluation (RQ3):** The proposed hybrid model is highly competitive, offering a distinct advantage in boundary detection. The best results were achieved with an **unweighted 4 multi-losses configuration during training**. Adaptive multi-losses configuration during training hypothesis after experiment are not outperforming the multi-ones.

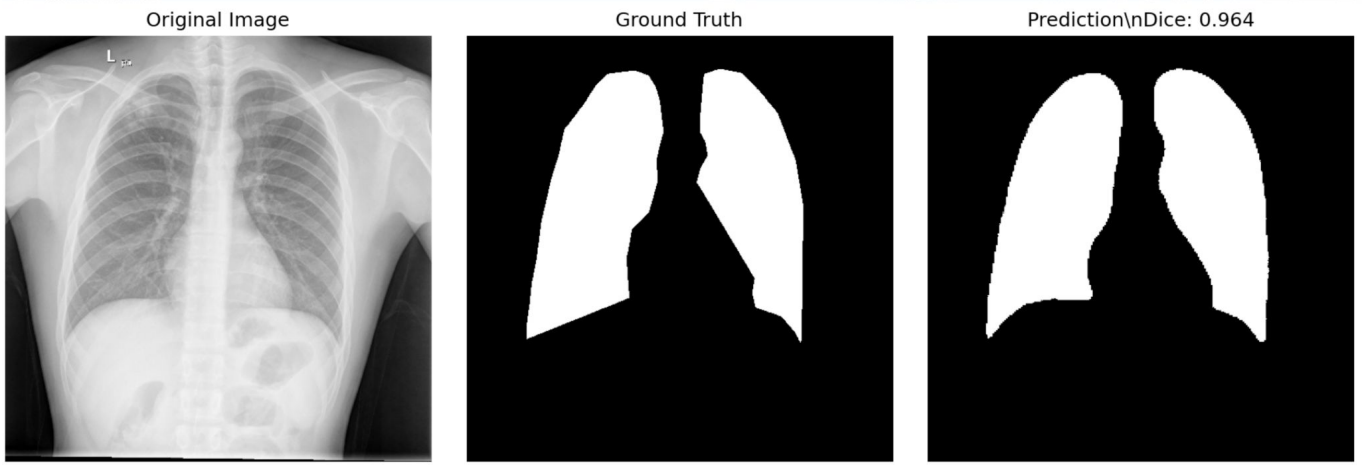
**Limitations**: Focus on architecture comparison; mathematical metrics may not reflect clinical impact; requires clinical validation

**Future Directions**: Model efficiency, clinical testing with radiologists, extended applications.

# Appendix

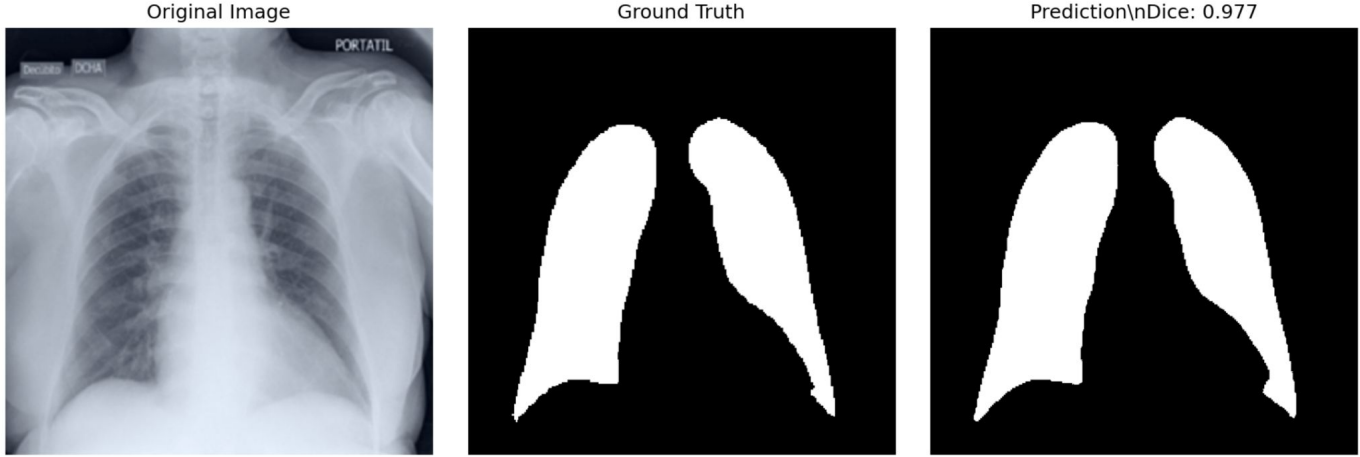
**Hybrid model**

**Test on unseen MC+SHZ image**



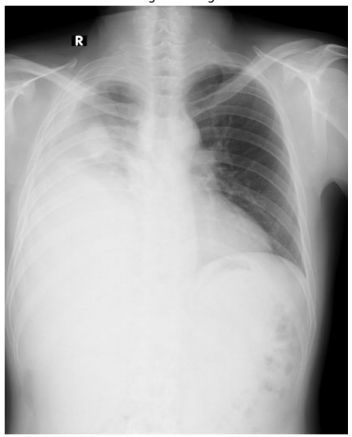

**Hybrid model**

**Test on unseen Q-COVID19 image**



## Hybrid Model test on real chest Xray Image from Vietnam, a dataset from VinDR

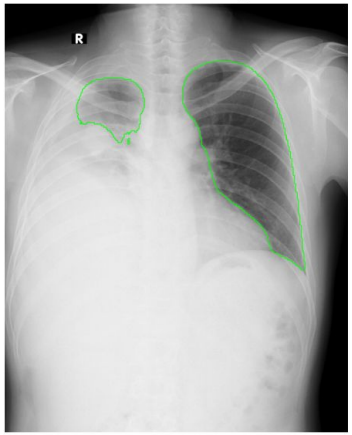
Original Image



Predicted Mask

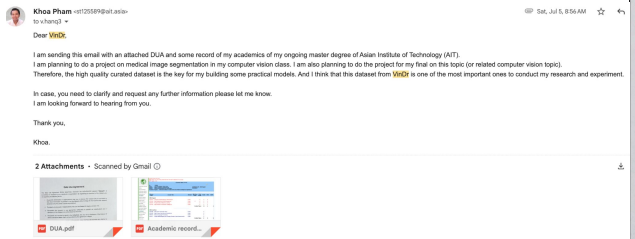


Green Contour Overlay



Red Filled Overlay





Khoa Pham <st120589@ait.asia> Sat, Jul 5, 8:56 AM

to v.hang1

Dear VinDr,

I am sending this email with an attached DUA and some record of my academics of my ongoing master degree of Asian Institute of Technology (AIT).
I am planning to do a project on medical image segmentation in my computer vision class. I am also planning to do the project for my final on this topic (or related computer vision topic).
Therefore, the high quality curated dataset is the key for my building some practical models. And I think that this dataset from VinDr is one of the most important ones to conduct my research and experiment.

In case, you need to clarify and request any further information please let me know.
I am looking forward to hearing from you.

Thank you,

Khoa.

2 Attachments • Scanned by Gmail

DUA.pdf

Academic record...

Nguyễn Quý Hà (VinBigdata_Managing Director/ Giám đốc Quản lý) <v.md@vinbigdata.org> Sat, Aug 2, 5:53 PM

to me

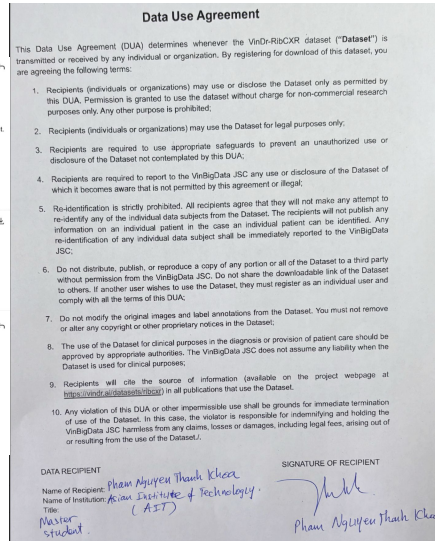
It looks like this message is in Vietnamese

Translate to English

Dear Khoa,

Sorry for the late response. Please request an access to the dataset via this link: https://drive.google.com/drive/folders/1EyardO9Mtt9VcNobtIP1_YlP5DI0B96I?usp=sharing

Thanks,
-Ha



Data Use Agreement

This Data Use Agreement (DUA) determines whenever the VinDr-RibCXR dataset ("**Dataset**") is transmitted or received by any individual or organization. By registering for download of this dataset, you are agreeing the following terms:

1. Recipients (individuals or organizations) may use or disclose the Dataset only as permitted by this DUA. Permission is granted to use the dataset without charge for non-commercial research purposes only. Any other purpose is prohibited;
2. Recipients (individuals or organizations) may use the Dataset for legal purposes only;
3. Recipients are required to use appropriate safeguards to prevent an unauthorized use or disclosure of the Dataset not contemplated by this DUA;
4. Recipients are required to report to the VinBigData JSC any use or disclosure of the Dataset of which it becomes aware that is not permitted by this agreement or illegal;
5. Re-identification is strictly prohibited. All recipients agree that they will not make any attempt to re-identify any of the individual data subjects from the Dataset. The recipients will not publish any information on an individual patient in the case an individual patient can be identified. Any re-identification of any individual data subject shall be immediately reported to the VinBigData JSC;
6. Do not distribute, publish, or reproduce a copy of any portion or all of the Dataset to a third party without permission from the VinBigData JSC; Do not share the downloadable link of the Dataset to others. If another user wishes to use the Dataset, they must register as an individual user and comply with all the terms of this DUA;
7. Do not modify the original images and label annotations from the Dataset. You must not remove or alter any copyright or other proprietary notices in the Dataset;
8. The use of the Dataset for clinical purposes in the diagnosis or provision of patient care should be approved by appropriate authorities. The VinBigData JSC does not assume any liability when the Dataset is used for clinical purposes;
9. Recipients will cite the source of information (available on the project webpage at https://vindr.ai/datasets/ribcxr) in all publications that use the Dataset.
10. Any violation of this DUA or other impermissible use shall be grounds for immediate termination of use of the Dataset. In this case, the violator is responsible for indemnifying and holding the VinBigData JSC harmless from any claims, losses or damages, including legal fees, arising out of or resulting from the use of the Dataset.;

DATA RECIPIENT

Name of Recipient: Pham Nguyen Thanh Khoa

Name of Institution: Asian Institute of Technology (AIT)

Title: Master student

SIGNATURE OF RECIPIENT

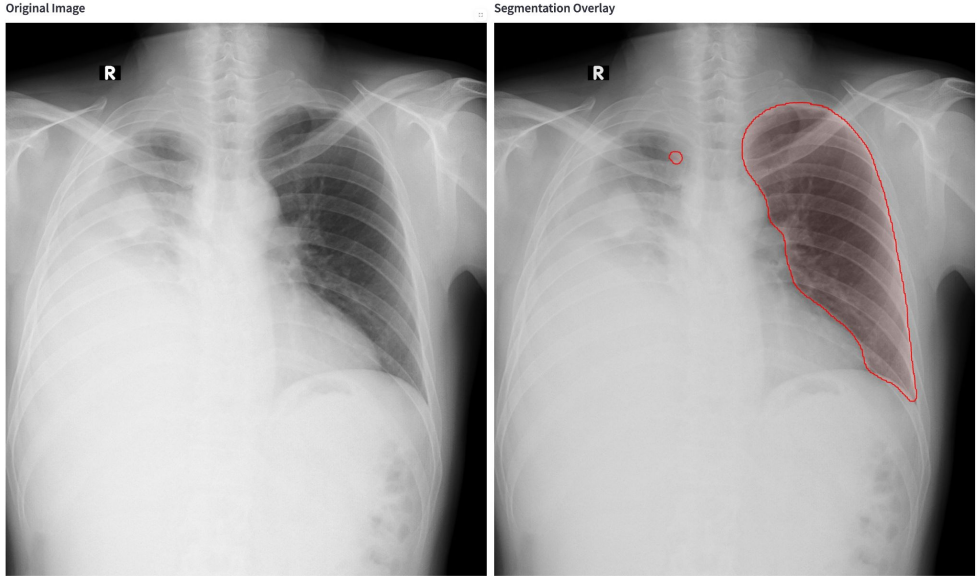
Pham Nguyen Thanh Kho

```
edictor_path = '/content/drive/MyDrive/AIT_LungSegmentation/saved_models_advanced/lung_segmentation_inference_ready_20250927_155222.pth'
st_image_path = "/content/drive/MyDrive/AIT_LungSegmentation/data/VinDr_Xray/VinDr_RibCXR_val_011.png"

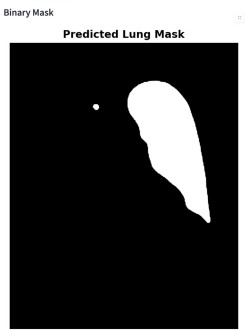
Simple test that handles size mismatches automatically
ntour_overlay, filled_overlay = simple_overlay_test(predictor_path, test_image_path)

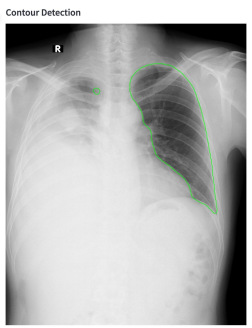
Loading image and running prediction...
Debugging image and mask sizes:
 Original image shape: (2880, 2304, 3)
 Mask shape: (512, 512)
 Probability mask shape: (512, 512)
 Original image dtype: uint8
 Mask dtype: float32
 Mask value range: 0.000 to 1.000
 Prob mask dtype: float32
 Prob mask value range: 0.012 to 0.998
 Creating overlays...
 Size mismatch detected: Image (2880, 2304) vs Mask (512, 512)
sizing mask to match image size...
 Size mismatch detected: Image (2880, 2304) vs Mask (512, 512)
```

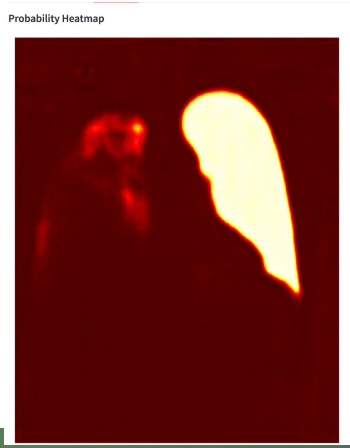
## CNN Only

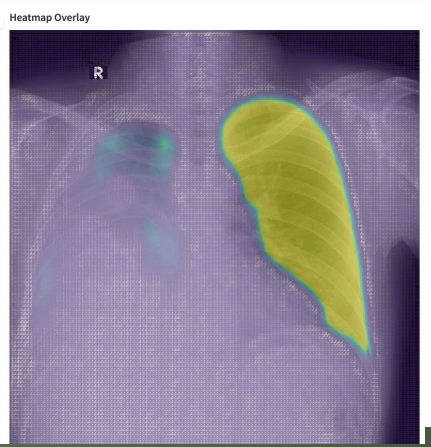

Original Image
R


Segmentation Overlay
R


Binary Mask
Predicted Lung Mask

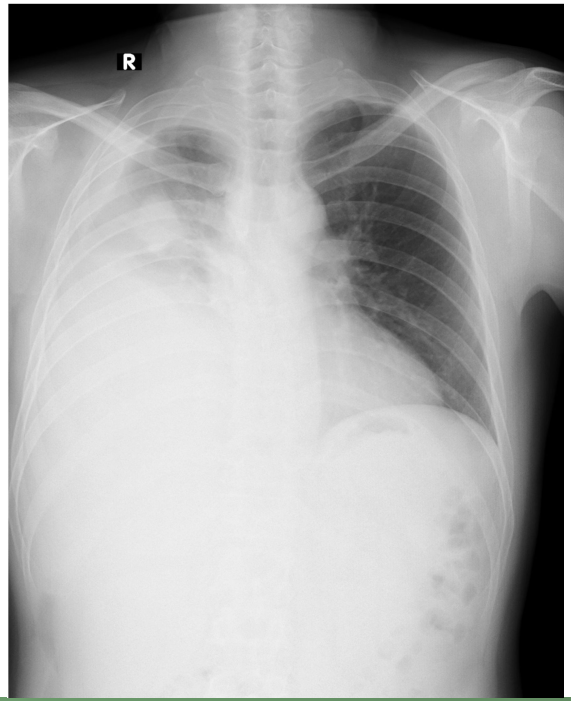

Contour Detection
R

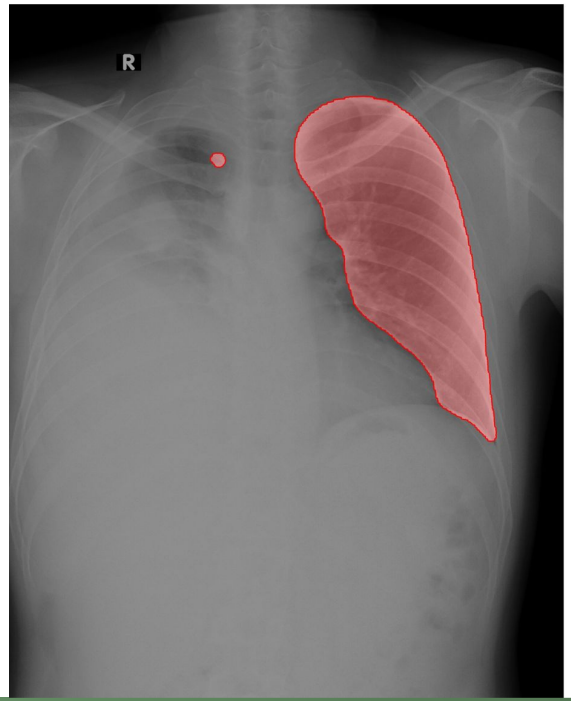

Probability Heatmap
1.0
0.8
0.6
0.4
0.2
0.0
Probability


Heatmap Overlay
R

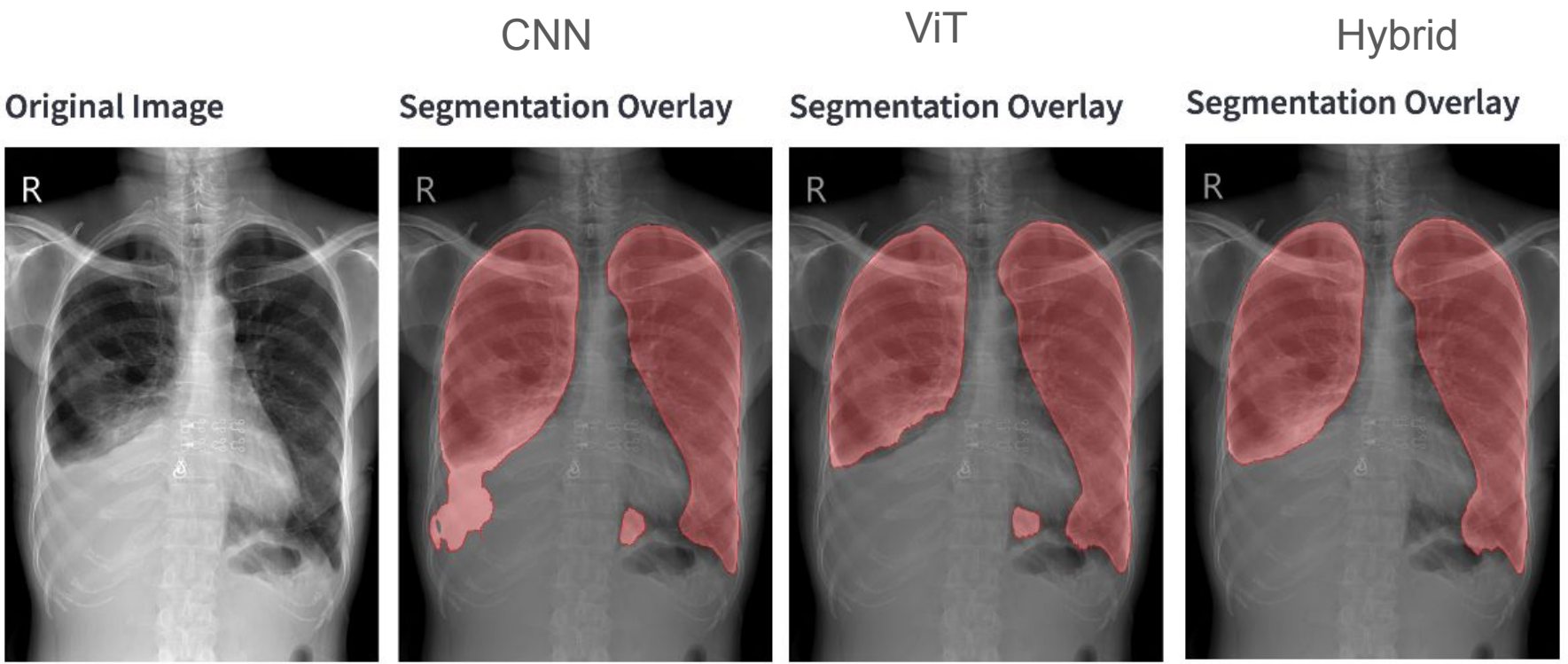
CNN

Hybrid







CNN
ViT
Hybrid
Original Image
Segmentation Overlay
Segmentation Overlay
Segmentation Overlay
R
R
R
R